



AKTU

B. Tech II-Year



Digital Electronics

Unit-1 in One Shot

Pdf Notes + AKTU PYQs + Imp. Questions



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DIGITAL ELECTRONICS

BOE310/ BOE410/BOE310H/ BOE410H

UNIT-I Syllabus

Digital System And Binary Numbers: Number System and its arithmetic Signed binary numbers, Logic simplification and combinational logic design: Binary codes, code conversion, review of Boolean algebra and Demorgans theorem, SOP & POS forms, Canonical forms, Karnaugh maps method up to five variable, Don't care conditions, POS simplification, NAND and NOR implementation, Quine McClusky method (Tabular method)

In digital technology ,the most commonly used number systems are

(1) Binary

(2) Octal

(3) Decimal

(4) Hexadecimal

Number System	Base / Radix	Digits Used	Example
Binary	2	0, 1	$(101111)_2$
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	$(47525)_8$
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	$(316875)_{10}$
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	$(9B21A)_{16}$

NOTE : A = 10, B = 11, C = 12, D = 13, E = 14, F = 15

(1) Binary to decimal : Example $(1011.11)_2 = (?)_{10}$

$$\begin{aligned}
 (1011.11)_2 &= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (1 \times 2^{-2}) \\
 &= 8 + 0 + 2 + 1 + 0.5 + 0.25 \\
 &= (11.75)_{10} \quad \underline{\text{Ans}}
 \end{aligned}$$

(2) Octal to decimal : Example $(475.25)_8 = (?)_{10}$

$$\begin{aligned}
 (475.25)_8 &= (4 \times 8^2) + (7 \times 8^1) + (5 \times 8^0) + (2 \times 8^{-1}) + (5 \times 8^{-2}) \\
 &= 256 + 56 + 5 + 0.25 + 0.0781 \\
 &= (317.3281)_{10} \quad \underline{\text{Ans}}
 \end{aligned}$$

To decimal
multiply by powers
other than 10

From Decimal
divide by base
other than 10

(3) Hexadecimal to decimal : Example $(9B2.1A)_{16} = (?)_{10}$

$$\begin{aligned}(9B2.1A)_{16} &= (9 \times 16^2) + (B \times 16^1) + (2 \times 16^0) + (1 \times 16^{-1}) + (A \times 16^{-2}) \\&= (9 \times 16^2) + (11 \times 16^1) + (2 \times 16^0) + (1 \times 16^{-1}) + (10 \times 16^{-2}) \\&= (2482.10156)_{10} \quad \underline{\text{Ans.}}\end{aligned}$$

Other conversion : Example $(3102.12)_4 = (?)_{10}$

$$\begin{aligned}(3102.12)_4 &= (3 \times 4^3) + (1 \times 4^2) + (0 \times 4^1) + (2 \times 4^0) + (1 \times 4^{-1}) + (2 \times 4^{-2}) \\&= 192 + 16 + 0 + 2 + 0.25 + 0.125 \\&= (210.375)_{10}\end{aligned}$$

(4) Decimal to Binary

Example : $(319.6875)_{10} = (?)_2$

2	319	R
2	159	1
2	79	1
2	39	1
2	19	1
2	9	1
2	4	0
2	2	0
	1	

Four
times

$$0.6875 \times 2 = 1.3750$$

$$0.3750 \times 2 = 0.7500$$

$$0.7500 \times 2 = 1.50$$

$$0.50 \times 2 = 1.00$$

1
0
1
1

$(10011111.1011)_2$ Ans.

(5) **Decimal to Octal** : Example $(319.6875)_{10} = (?)_8$

8	319	R 7
8	39	7
	4	

$$\begin{array}{rcl}
 0.6875 \times 8 & = & 5.5 \quad 5 \\
 0.5 \times 8 & = & 4.0 \quad 4 \\
 0.0 \times 8 & = & 0.0 \quad 0 \\
 0.0 \times 8 & = & 0.0 \quad 0
 \end{array}$$

$(477.5400)_8$ Ans

(6) **Decimal to Hexadecimal** : Example $(319.6875)_{10} = (?)_{16}$

16	319	R 15 → F
16	19	3
	1	

$$\begin{array}{rcl}
 0.6875 \times 16 & = & 11.0000 \quad 11 \rightarrow B \\
 0.0000 \times 16 & = & 0.0 \quad 0 \\
 0.0 \times 16 & = & 0.0 \quad 0 \\
 0.0 \times 16 & = & 0.0 \quad 0
 \end{array}$$

$(13F.B)_{16}$ Ans

Other conversion : Example $(319.6875)_{10} = (?)_4$

4	319	R
4	79	3
4	19	3
4	4	3
	1	0

$$0.6875 \times 4 = 2.75$$

$$0.75 \times 4 = 3.0$$

$$0.0 \times 4 = 0.0$$

$$0.0 \times 4 = 0.0$$

2
3
0
0

$$(10333.23)_4$$

Ans.

(7) Binary to octal conversion

Octal 8 = 2³
← 3 3 →

Example : $(100\ 111\ 111.1011)_2 = (?)_8$

100 111 111 , 101 100

4 7 7 . 5 4

$(477.54)_8$ Ans

(8) Octal to binary conversion

Example : $(325.63)_8 = (?)_2$

3 2 5 . 6 3
↓
011 010 101 . 110 011

$(011010101.110011)_2$ Ans

(9) Binary to Hexadecimal conversion

Example : $(100\ 111111.1011)_2 = (?)_{16}$

Hexadecimal 16 = 2⁴
← 4 4 →

0001 0011 1111 . 1011
↓
1 3 15 . 11₂
F B
 $(13F.B)_{16}$

(10) Hexadecimal to binary conversion

Example : $(5A9.B4)_{16} = (?)_2$

5 A 9 . B 4
↓
0101 1010 1001 . 1011 0100

$(010110101001.10110100)_2$ Ans

(11) Octal to Hexadecimal conversion

3 → 4

Easy Method : Octal → Binary → Hexadecimal

Example : $(745)_8 = (?)_{16}$

Step 1 Octal to Binary

$\begin{matrix} 7 & 4 & 5 \\ \rightarrow & 111 & 100 & 101 \end{matrix}$

$(745)_8 = (111100101)_2$

Step 2 Binary to Hexadecimal

$\begin{matrix} \boxed{0001} & \boxed{1110} & \boxed{0101} \\ \rightarrow & 1 & E & 5 \end{matrix}$

$(1E5)_{16}$ Ans

(12) Hexadecimal to octal

4 → 3

Easy Method : Hexadecimal → Binary → Octal

Example : $(13F.B)_{16} = (?)_8$ Don't read it as 13.

Step 1 Hexadecimal to Binary

$\begin{matrix} 1 & 3 & F & . & B \\ \rightarrow & 0001 & 0011 & 1111 & . & 1011 \end{matrix}$

Step 2 Binary to Octal

$\begin{matrix} \boxed{100} & \boxed{111} & \boxed{111} & . & \boxed{101} & \boxed{100} \\ \rightarrow & 4 & 7 & 7 & . & 5 & 4 \end{matrix}$

$(477.54)_8$ Ans

Other base to Other base Conversion

➤ Universal Method of Number system conversion



Example $(432)_5 = (?)_9$

step 1

$$(4 \times 5^2) + (3 \times 5^1) + (2 \times 5^0)$$

$$(4 \times 25) + 15 + 2$$

$$100 + 15 + 2$$

$$(117)_{10}$$

9	117	0
9	13	4
	1	

$$(140)_9 \text{ Ans}$$

Other conversion : Example $(1011)_2 = (?)_4$

$$\begin{aligned}
 &= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \\
 &= 8 + 0 + 2 + 1 \\
 &= (11)_{10}
 \end{aligned}$$

4	11	3	
	2	3	↗
	$(23)_4$		<u>Ans</u>

Other conversion : Example $(637)_9 = (?)_5$

$$\begin{aligned}
 &= (6 \times 9^2) + (3 \times 9^1) + (7 \times 9^0) \\
 &= (6 \times 81) + 27 + 7 \\
 &= (520)_{10}
 \end{aligned}$$

5	520	0	
5	104	4	
5	20	0	
	4		↖
$(4040)_5$		<u>Ans</u>	

To find base x

Example : Determine the values of base if $(193)_x = (623)_8$

$(193)_x \rightarrow (\quad)_{10} \leftarrow (623)_8$, then Equate

$$(1 \times x^2) + (9 \times x^1) + (3 \times x^0) = (6 \times 8^2) + (2 \times 8^1) + (3 \times 8^0)$$

$$x^2 + 9x + 3 = 403$$

$$x^2 + 9x - 400 = 0$$

$$x^2 + 25x - 16x - 400 = 0$$

$$x(x + 25) - 16(x + 25) = 0$$

∴ Base cannot be -ve
 $x = -25$ X

$$x = 16$$

Que) Represent the decimal number 45 in binary and octal number system.

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(i) $(45)_{10} \rightarrow (?)_2$

2	45	^R 1
2	22	0
2	11	1
2	5	1
2	2	0
	1	

$(101101)_2$ Ans

(ii) $(45)_{10} \rightarrow (?)_8$

8	45	^R 5
	5	

$(55)_8$ Ans

Que) Convert the following numbers to their respective bases:

- (i) $(47.625)_{10}$ to binary
- (ii) $(101101.101)_2$ to decimal
- (iii) $(3F9)_{16}$ to decimal
- (iv) $(543)_8$ to binary

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(i) $(47.625)_{10} \rightarrow (?)_2$

2	47	1
2	23	1
2	11	1
2	5	1
2	2	0
	1	

3 times

$$\left\{ \begin{array}{l} .625 \times 2 = 1.25 \quad 1 \\ .25 \times 2 = 0.50 \quad 0 \\ .50 \times 2 = 1.00 \quad 1 \end{array} \right.$$

$(101111.101)_2$

Ans

(ii) $(101101.101)_2 \rightarrow (?)_{10}$

$$(1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$$

$$= 32 + 8 + 4 + 1 + 0.5 + 0.125$$

$= (45.625)_{10}$ Ans

$$(iii) (3F9)_{16} \rightarrow (?)_{10}$$

$$= (3 \times 16^2) + (F \times 16^1) + (9 \times 16^0)$$

$$= (3 \times 16^2) + (15 \times 16^1) + (9 \times 16^0)$$

$$= 768 + 240 + 9$$

$$= 1017$$

$$(1017)_{10} \text{ Ans}$$

$$(iv) (543)_8 \rightarrow (?)_2$$

$$\begin{array}{ccc} 5 & 4 & 3 \\ \downarrow & \downarrow & \downarrow \\ 101 & 100 & 011 \end{array}$$

$$(101100011)_2 \text{ Ans}$$

Que) Convert hexadecimal number 3F to binary.

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$$(3F)_{16} \rightarrow (?)_2$$

$$\begin{array}{cc} 3 & 15 \\ \downarrow & \downarrow \\ 0011 & 1111 \end{array}$$

$$(111111)_2 \quad \underline{\underline{\text{Ans}}}$$

Gateway Classes

Que) Convert the following

(i) $(5162)_{10} = ()_2$

(ii) $(11011001)_2 = ()_{10}$

(iii) $(6273)_{10} = ()_8$

(iv) $(7860)_{10} = ()_{16}$

(v) $(A23B8)_{16} = ()_{10}$

(u) $(11011001)_2 \rightarrow ()_{10}$

$$(1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$128 + 64 + 16 + 8 + 1$$

$(217)_{10}$ Ans

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(i) $(5162)_{10} \rightarrow (?)_2$

2	5162	0
2	2581	1
2	1290	0
2	645	1
2	322	0
2	161	1
2	80	0
2	40	0
2	20	0
2	10	0
2	5	1
2	2	0
	1	

$(1010000101010)_2$

Ans

$$(iii) (6273)_{10} \rightarrow (?)_8$$

		R
8	6273	1
8	784	0
8	98	2
8	12	4
	1	

$$(14201)_8 \text{ Ans}$$

$$(iv) (7860)_{10} \rightarrow (?)_{16}$$

		R
16	7860	4
16	491	11
16	30	14
	1	

1 14 11 4

$$(1EB4)_{16} \text{ Ans}$$

$$(v) (A23B8)_{16} \rightarrow (?)_{10}$$



$$= (A \times 16^4) + (2 \times 16^3) + (3 \times 16^2) + (B \times 16^1) + (8 \times 16^0)$$

$$= (10 \times 16^4) + (2 \times 16^3) + (3 \times 16^2) + (11 \times 16^1) + (8 \times 16^0)$$

$$= 655360 + 8192 + 768 + 176 + 8$$

$$= 664504$$

$$(664504)_{10} \text{ Ans}$$

Que) How are binary digits used to express the integer and fractional parts of a number ?

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A number has two parts $\underline{\text{I}} \cdot \underline{\text{F}}$

1) Integer part (Left of the binary point)

The positions to the left of the binary point represent positive powers of 2

..... 2^4 2^3 2^2 2^1 2^0

2) Fractional part (Right of the binary point)

The positions to the right of the binary point represent negative powers of 2

2^{-1} 2^{-2} 2^{-3}

eg 1101
 $\hookrightarrow (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$
 13

eg 0.101
 $(1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$
 0.625

(1) Binary Addition

SUM	CARRY
$0 + 0 = 0$	0
$0 + 1 = 1$	0
$1 + 0 = 1$	0
$1 + 1 = 0$	1
$1 + 1 + 1 = 1$	1

Note:

(i) $1 + 1 = 10$

SUM = 0

Carry = 1

(ii) $1 + 1 + 1 = 11$

SUM = 1

Carry = 1

Example - 1 : Add $(1110)_2$ and $(1100)_2$

$$\begin{array}{r}
 \\
 \\
 + \\
 \hline
 \\
 \hline

 \end{array}$$

Example -2 : Add $(11001)_2$ and $(10111)_2$

$$\begin{array}{r}
 \\
 \\
 + \\
 \hline
 \\
 \hline

 \end{array}$$

Example - 3 : Add binary number $(1110.110)_2$ to $(1100.011)_2$

$$\begin{array}{r}
 \\
 \\
 + \\
 \hline
 \\
 \hline

 \end{array}$$

Binary Arithmetic

(2) Binary Subtraction

Difference	Borrow
$0 - 0 = 0$	0
$1 - 0 = 1$	0
$1 - 1 = 0$	0
$0 - 1 = 1$	1

Example -1 : Subtract $(10110)_2$ from $(11011)_2$

$$\begin{array}{r}
 1-1=0 \\
 \begin{array}{r}
 11011 \\
 -10110 \\
 \hline
 00101
 \end{array}
 \end{array}$$

$$\begin{array}{l}
 1-1=0 \\
 (B) \\
 0-0=0
 \end{array}$$

Example - 2 : Subtract $(00001)_2$ from $(11000)_2$

$$\begin{array}{r}
 1-1=0 \\
 \begin{array}{r}
 11000 \\
 -00001 \\
 \hline
 10111
 \end{array}
 \end{array}$$

Example -3 : Subtract $(0101)_2$ from $(1100)_2$

step 1

$$\begin{array}{r}
 1-1=0 \\
 \begin{array}{r}
 1100 \\
 -0101 \\
 \hline
 11
 \end{array}
 \end{array}$$

step 2

$$\begin{array}{r}
 1-1=0 \\
 \begin{array}{r}
 1100 \\
 -0101 \\
 \hline
 0111
 \end{array}
 \end{array}$$

Example - 4 : Subtract $(010111)_2$ from $(100010)_2$

$$\begin{array}{r}
 \begin{array}{r}
 01110 \\
 100010 \\
 -010111 \\
 \hline
 001011
 \end{array}
 \end{array}$$

Complements

Complements are used in the digital computers in order to simplify the subtraction operation and for the logical manipulation.

For Binary System

- (i) 1' S complement
- (ii) 2 ' S complement

1' S complement (0 $\rightarrow$ 1 , 1 $\rightarrow$ 0)

Given Number

1 0 1 0

1 1 0 1 0 1 0 1

1 0 0 1

1' S complement

0 1 0 1

0 0 1 0 1 0 1 0

0 1 1 0

2' S complement

$$2' \text{ S complement} = 1' \text{ S complement} + 1$$

→ on extreme right of the number

Given

1 0 1 1

1' S complement

0 1 0 0

2' S complement

$$\begin{array}{r} 0100 \\ + 1 \\ \hline 0101 \end{array}$$

1 0 1 1 0 0 1 0

0 1 0 0 1 1 0 1

$$\begin{array}{r} 01001101 \\ + 1 \\ \hline 01001110 \end{array}$$

Que) What is 1's complement and how is it different from 2's complement ?

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1) 1's complement (0 $\rightarrow$ 1, 1 $\rightarrow$ 0)

2's complement = 1's complement + 1

2) Representation of positive and negative number using 1's complement

Given

+5

1's complement

0 1 0 1
+ve

-5

1 0 1 0
-ve

is 0 1 0 1
1 0 1 0

Representation of positive and negative number using 2's complement

Given

+5

2's complement

0 1 0 1
+ve

-5

1 0 1 1
-ve

is 0 1 0 1
1 0 1 0
+ 1
1 0 1 1

Complement Subtraction

$$\begin{array}{r}
 (\quad) - (\quad) \\
 \downarrow \\
 1's \text{ or } 2's \text{ or } 9's \text{ or } 10's \\
 + \quad (\quad)
 \end{array}$$

Make digits equal of both the numbers by appending 0 on extreme left of smaller number.

$$\begin{array}{l}
 (\quad) - (\quad) \rightarrow (\quad) - (0 \quad) \\
 \text{or} \\
 (\quad) - (\quad) \rightarrow (0 \quad) - (\quad)
 \end{array}$$

	Carry comes	Carry does not come
1's	Add carry to remaining ans to get final ans.	Ans is -ve , Take 1's complement of ans to get final ans.
2's	Neglect Carry, remaining ans is the final ans	Ans is -ve , Take 2's complement of ans to get final ans.
9's	Add carry to remaining ans to get final ans.	Ans is -ve , Take 9's complement of ans to get final ans.
10's	Subtract smallestdigit number from ans to get final ans	Ans is -ve , Take 10's complement of ans to get final ans.

Binary Subtraction using 1's complement

Example – 1 : Subtract using 1's complement $(1111)_2 - (1010)_2$

$$\begin{array}{r}
 \\
 + 1111 \\
 \hline
 10101
 \end{array}$$

1's complement of 1010 is 0101

$$\begin{array}{r}
 10101 \\
 + 0101 \\
 \hline
 11010
 \end{array}$$

carry ← 1

$$\begin{array}{r}
 0100 \\
 + 1 \\
 \hline
 0101
 \end{array}$$

Ans

$$\begin{array}{r}
 1111 \rightarrow 15 \\
 1010 \rightarrow -10 \\
 \hline
 5 \\
 0101
 \end{array}$$

Example – 2 : Subtract using 1's complement $(10111)_2 - (110011)_2$

$$(010111) - (110011)$$

$$\begin{array}{r}
 001100 \\
 + 010111 \\
 \hline
 100011
 \end{array}$$

No carry - Ans is -ve
1's complement of 100011 is 011100

$$\rightarrow -(011100) \text{ Ans }$$

$$\begin{array}{r}
 10111 \rightarrow 23 \\
 110011 \rightarrow -51 \\
 \hline
 -28 \\
 2 \\
 -(011100)
 \end{array}$$

Example – 1 : Subtract using 2's complement

$$(1001)_2 - (0100)_2$$

1's
2's

$$\begin{array}{r} 1011 \\ + 1 \\ \hline 1100 \end{array}$$

$$\begin{array}{r} 1001 \\ + 1100 \\ \hline 10101 \end{array}$$

Carry

Neglect carry (0101) Ans.

$$\begin{array}{r} 9 \\ - 5 \\ \hline 4 \end{array}$$

Example – 2 : Subtract using 2's complement $(010110)_2 - (100101)_2$

$$\begin{array}{r} 011010 \\ + 1 \\ \hline 011011 \end{array}$$

+

$$\begin{array}{r} 010110 \\ + 011011 \\ \hline 110001 \end{array}$$

→

No carry

Ans is -ve

$$110001$$

$$\begin{array}{r} 001110 \\ + 1 \\ \hline 001111 \end{array}$$

-(001111) Ans

$$\begin{array}{r} 22 \\ - 37 \\ \hline -15 \end{array}$$

Example — 3 : Subtract using 2' s complement $(111)_2 - (1010)_2$

$$(0111)_2 - (1010)_2$$

$$\begin{array}{r} 0101 \\ +1 \\ \hline 0110 \end{array}$$

$$\begin{array}{r} 0111 \\ + 0110 \\ \hline \end{array}$$

1101

→ No carry

Ans is -ve

$$\begin{array}{r} 1101 \\ 0010 \\ +1 \\ \hline 0011 \end{array}$$

-(0011)

Ans

$$\begin{array}{r} 7 \\ -10 \\ \hline -3 \\ \hline \end{array}$$

Que) Evaluate $(1011)_2 - (1101)_2$ using 1's and 2's complement method.

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(1) 1's complement method

$$\begin{array}{r} (1011) - (1101) \\ \text{is } \rightarrow 0010 \end{array}$$

$$\begin{array}{r} 1011 \\ + 0010 \\ \hline 1101 \end{array}$$

No carry ans is -ve
is $\rightarrow 0010$ } $\rightarrow -(0010)$
Ans

(2) 2's complement method

$$\begin{array}{r} (1011) - (1101) \\ \text{is } \rightarrow 0010 \\ + 1 \\ \hline 0011 \end{array}$$

$$\begin{array}{r} 1011 \\ + 0011 \\ \hline 1110 \end{array}$$

No carry ans is -ve
2's $\rightarrow 0001$
 $+ 1$
 $\hline 0010$ } $-(0010)$
Ans

(i) 9' S complement

(ii) 10' S complement

9' S complement

Given Number

2467

9' S complement

9999

-2467

7532

→ Greatest four digit number

∴ 2467 is a four digit number

75468

99999

-75468

24531

→ Greatest five digit number

∴ 75468 is a five digit number

eg

$$\begin{array}{r} 7 \\ 9 \\ - 7 \\ \hline 2 \end{array}$$

eg

$$\begin{array}{r} 23 \\ 99 \\ - 23 \\ \hline 76 \end{array}$$

$$\underline{10' \text{ S complement}} = 9' \text{ S complement} + 1$$

Given Number

9' S complement

10' S complement

4862

5137

$$(9999 - 4862)$$

5137

+ 1

5138

is 10's

complement of
4862

65432

34567

$$(99999 - 65432)$$

34567

+ 1

34568

is 10's

complement of
65432

Binary Subtraction using 9's complement

Example – 1 : Subtract 234 from 678 using 9's complement subtraction.

$$\begin{array}{r}
 678 - 234 \\
 \begin{array}{r}
 999 \\
 - 234 \\
 \hline
 765
 \end{array} \\
 \swarrow + \searrow \\
 \begin{array}{r}
 678 \\
 + 765 \\
 \hline
 1443
 \end{array}
 \end{array}$$

Carry →

$$\begin{array}{r}
 443 \\
 + 1 \\
 \hline
 444 \quad \text{Ans}
 \end{array}$$

Example – 1 : Subtract 678 from 234 using 9's complement subtraction.

$$\begin{array}{r}
 234 - 678 \\
 \begin{array}{r}
 999 \\
 - 678 \\
 \hline
 321
 \end{array} \\
 \swarrow + \searrow \\
 \begin{array}{r}
 234 \\
 + 321 \\
 \hline
 555
 \end{array}
 \end{array}$$

9's

$$\begin{array}{r}
 999 \\
 - 555 \\
 \hline
 444
 \end{array}$$

No carry ans is -ve } $-(444)$ Ans

Binary Subtraction using 10's complement

Example – 1 : Subtract $(9754)_{10} - (364)_{10}$ using 10's complement subtraction.

$$\begin{array}{r} 9754 \\ + 626 \\ \hline 10380 \end{array}$$

$$\begin{array}{r} 999 \\ - 364 \\ \hline 625 \\ + 1 \\ \hline 626 \end{array}$$

9's
10's

$$\begin{array}{r} 10380 \\ - 1000 \\ \hline 9380 \end{array} \quad \text{Ans}$$

9754
is a 4 digit number
smallest 4 digit number
1000 is subtracted from ans to get final answer.

Example – 1 : Subtract $(670)_{10} - (831)_{10}$ using 10's complement subtraction

$$\begin{array}{r} 670 \\ + 169 \\ \hline 839 \end{array}$$

$$\begin{array}{r} 999 \\ - 831 \\ \hline 168 \\ + 1 \\ \hline 169 \end{array}$$

9's
10's

No carry : ans is -ve

$$\begin{array}{r} 839 \\ 10's \rightarrow \begin{array}{r} 999 \\ - 839 \\ \hline 160 \\ + 1 \\ \hline 161 \end{array} \end{array}$$

– (161) Ans

Signed binary numbers are used to represent both positive and negative integers in computing, typically using the MSB(Most Significant Bit) as a sign bit : 0 for positive and 1 for negative.

The three main representation methods are :-

Sign –Magnitude : The MSB is the sign , and the remaining n-1 bits represent the magnitude.

For example

4 bit +3 is 0011

4 bit -3 is 1011

3 bit 011

3 bit 011

0011 → +ve

1011 → -ve

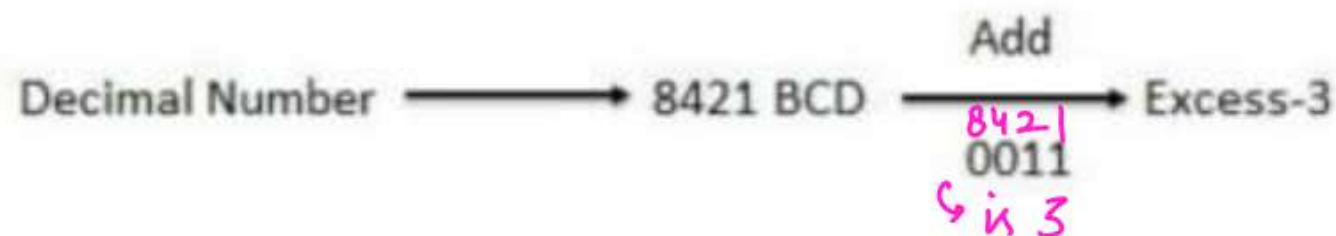
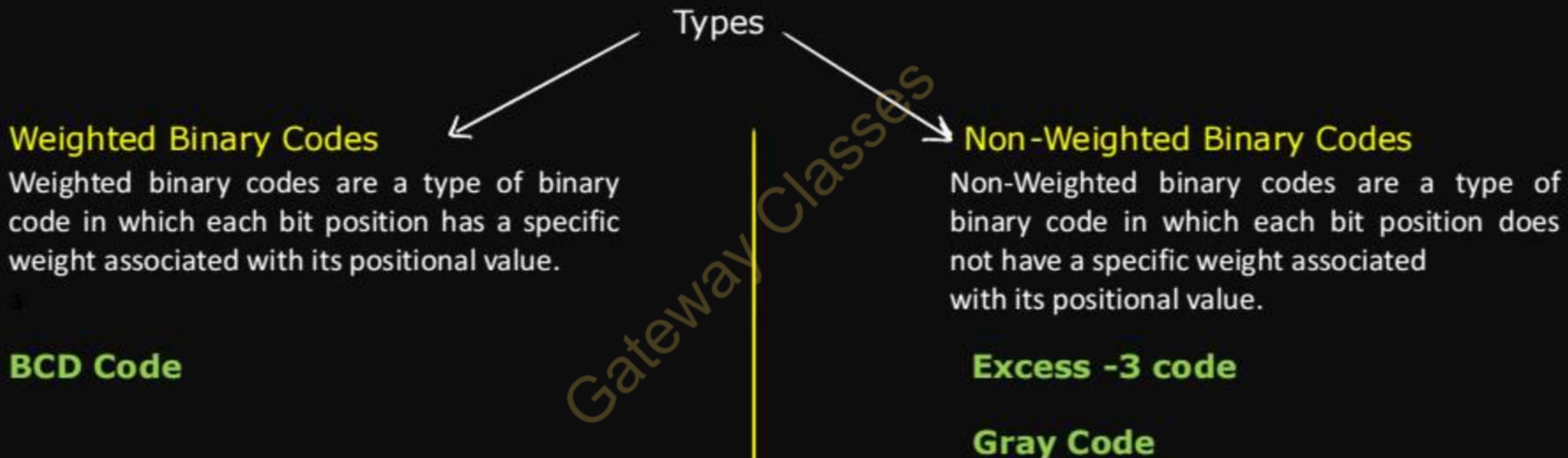
1's Complement 1 → 0 , 0 → 1

2's Complement 1's complement + 1

Que) Define the term binary codes with an example.

AKTU PYQ-2021-22

A binary code is a type of digital code consisting of two digits, 0 and 1.



⁽¹⁾
Que) Define BCD codes and convert $(A5D8)_{16}$ into BCD number.

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Binary Coded Decimal (BCD) is a binary encoding system in which each decimal digit is represented by a fixed number of binary bits, typically **four**.

In BCD, each decimal digit (0-9) is converted into its 4-bit binary equivalent. For example:

- Decimal 0 is represented as 0000 in BCD.
- Decimal 1 is represented as 0001 in BCD.
- Decimal 2 is represented as 0010 in BCD and so on.

0-9
≥ 10

Binary, BCD same

Write individual digit in binary of four.

For instance, the decimal number 57 would be represented in BCD as two separate 4-bit binary numbers:

- Decimal 5 becomes 0101 and Decimal 7 becomes 0111

So, 57 in BCD is represented as 0101 0111.

Convert $(A5D8)_{16}$ into BCD number.

$$(A5D8)_{16} \rightarrow BCD$$

$$(A5D8)_{16} \rightarrow \text{Decimal} \rightarrow BCD$$

(i)

$$\begin{aligned} A5D8 &\rightarrow (A \times 16^3) + (5 \times 16^2) + (D \times 16^1) + (8 \times 16^0) \\ &= (10 \times 16^3) + (5 \times 16^2) + (13 \times 16^1) + (8 \times 16^0) \\ &= 40960 + 1280 + 208 + 8 \\ &= 42456 \end{aligned}$$

$$(A5D8)_{16} = (42456)_{10}$$

$$(ii) (42456)_{10} \rightarrow BCD$$

$$4 \rightarrow 0100$$

$$2 \rightarrow 0010$$

$$4 \rightarrow 0100$$

$$5 \rightarrow 0101$$

$$6 \rightarrow 0110$$

$$\therefore (01000010010001010110)$$

Ans

Que) How BCD addition is carried out ?

If $\text{sum} \leq 9$ No correction needed
If $\text{sum} > 9$, add (0110) i.e. 6

eg

$$\begin{array}{r} 4 + 3 \\ 0100 \\ + 0011 \\ \hline 0111 \end{array}$$

→ 7 (valid BCD)

7 Ans.

eg $7 + 5$

$$\begin{array}{r} 0111 \\ + 0101 \\ \hline 1100 \end{array}$$

→ 12 (invalid BCD)
∴ add 0110

$$\begin{array}{r} 1100 \\ + 0110 \\ \hline 10010 \end{array}$$

0001 0010
→ 1 2

∴ 12 Ans

Que) What is Excess-3 code ? Convert $(57)_{10}$ to Excess-3 code and explain the advantages of Excess-3 code over BCD.

AKTU PYQ-2025-26, Sem III

Excess 3 number = BCD number + $(0011)_2$
 $\rightarrow 3$

Advantages:-

- Excess-3 is self-complementing

Taking 1's complement of an Excess-3 number directly gives the Excess-3 code of 9's complement of that decimal number.

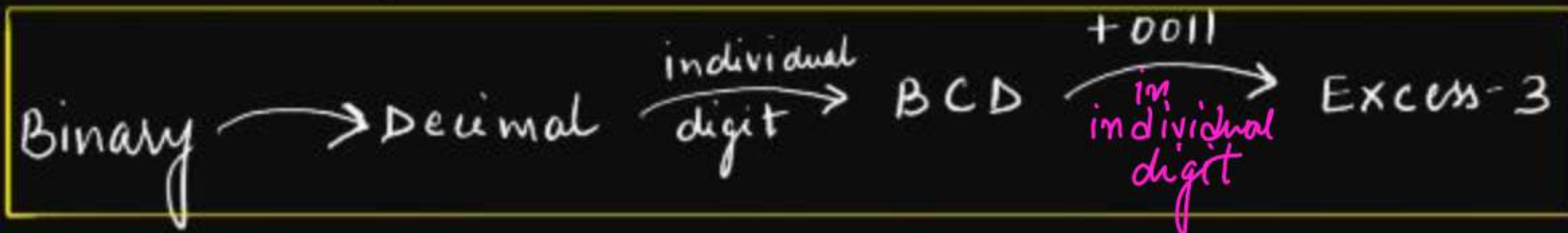
↓
 This makes decimal subtraction easier

BCD does not have this property

- Excess-3 has better Arithmetic simplicity

BCD INPUTS				EXCESS-3 OUTPUTS			
D3	D2	D1	D0	E3	E2	E1	E0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

Convert $(57)_{10}$ to Excess-3 code.



$$\begin{array}{lclcl} 5 & \xrightarrow{\text{BCD}} & 0101 & \xrightarrow{+0011} & 1000 \\ 7 & \xrightarrow{\text{BCD}} & 0111 & \xrightarrow{+0011} & 1010 \end{array}$$

$\therefore (10001010)$ Ans

Que) What is Gray code ? Convert $(1010)_2$ to Gray Code and explain its significance.

AKTU PYQ-2024-25

Concept

$B \rightarrow G$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$G_3 = B_3$

$G_2 = B_3 \text{ EX-OR } B_2 =$

$G_1 = B_2 \text{ EX-OR } B_1 =$

$G_0 = B_1 \text{ EX-OR } B_0 =$

Concept

$B \rightarrow G$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$G_3 = B_3$

$G_2 = B_3 \text{ EX-OR } B_2 =$

$G_1 = B_2 \text{ EX-OR } B_1 =$

$G_0 = B_1 \text{ EX-OR } B_0 =$

Que $(1010)_2 \rightarrow \text{Gray code}$

$B \rightarrow G$

$1010 \rightarrow ?$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$G_3 = 1$

$G_2 = 1 \text{ EX-OR } 0 = 1$

$G_1 = 0 \text{ EX-OR } 1 = 1$

$G_0 = 1 \text{ EX-OR } 0 = 1$

Ans **1111**

Que) Interpret the binary number $(1011)_2$ into (i) Gray code (ii) Excess-3 Code.

(i) $(1011)_2 \rightarrow$ Gray code

1 0 1 1
B3 B2 B1 B0

$$\begin{aligned} G_3 &= B_3 = 1 \\ G_2 &= B_3 \oplus B_2 = 1 \oplus 0 = 1 \\ G_1 &= B_2 \oplus B_1 = 0 \oplus 1 = 1 \\ G_0 &= B_1 \oplus B_0 = 1 \oplus 1 = 0 \end{aligned}$$

(1110) Ans

(ii) $(1011)_2 \rightarrow$ Excess-3 code

Binary $\rightarrow$ Decimal $\xrightarrow{\text{individual digit}}$ BCD $\xrightarrow{\text{+001 in individual digit}}$ Excess-3

Binary to Decimal

AKTU PYQ-2022-23

$(1011)_2 \rightarrow ()_{10}$

$$\begin{aligned} &= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \\ &= (11)_{10} \end{aligned}$$

Decimal to BCD

1 $\rightarrow$ 0001
1 $\rightarrow$ 0001

BCD to Excess-3

0001 $\xrightarrow{+0011}$ 0100
0001 $\xrightarrow{+0011}$ 0100

(01000100) Ans

Boolean Algebra

- Boolean Algebra is used to analyze and simplify digital (logic) circuits.
- Each variable in Boolean algebra has values either 0 (Low) or 1 (High).
- The basic operations used in Boolean algebra are:

$A \in \begin{matrix} 0 \\ \text{or} \\ 1 \end{matrix}$

$B \in \begin{matrix} 0 \\ \text{or} \\ 1 \end{matrix}$

(a) Logical addition

(b) Logical multiplication

(c) Complementation

➤ Laws of Boolean Algebra

(1) Commutative Law

(2) Associative Law

(3) Distributive Law

(4) AND Laws

(5) OR Laws

(6) Inversion Law

(7) Absorption Law

(8) De Morgan's Law

(1) Commutative Law

The commutative law states that changing the sequence of the variables does not have any effect on the output of a logic circuit.

$$(a) A \cdot B = B \cdot A$$

$$0 \cdot 1 = 1 \cdot 0$$

$$(b) A + B = B + A$$

$$0 + 1 = 1 + 0$$

(2) Associative Law

This law states that the order in which the logic operations are performed is irrelevant as their effect is the same.

$$(a) (A \cdot B) \cdot C = A \cdot (B \cdot C)$$

$$(b) (A + B) + C = A + (B + C)$$

(3) Distributive Law

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

(4) AND Laws

These laws use the AND operation. Therefore, they are called AND laws.

(a) $A \cdot 0 = 0$

A	0	$A \cdot 0$
0	0	0
1	0	0

(b) $A \cdot 1 = A$

A	1	$A \cdot 1$
0	1	0
1	1	1

(c) $A \cdot A = A$

A	A	$A \cdot A$
0	0	0
1	1	1

(d) $A \cdot \bar{A} = 0$

A	$\bar{A}$	$A \cdot \bar{A}$
0	1	0
1	0	0

(5) OR Laws

(a) $A + 0 = A$

A	0	$A + 0$
0	0	0
1	0	1

(b) $A + 1 = 1$

A	1	$A + 1$
0	1	1
1	1	1

(c) $A + A = A$

A	A	$A + A$
0	0	0
1	1	1

(d) $A + \bar{A} = 1$

A	$\bar{A}$	$A + \bar{A}$
0	1	1
1	0	1

(6) Inversion Law

The inversion law states that double inversion of a variable results in the original variable itself

$$\bar{\bar{A}} = A$$

A	$\bar{A}$	$\bar{\bar{A}}$
0	1	0
1	0	1

(7) Absorption Law

This law enables a reduction in a complicated expression to a simpler one by absorbing like terms.

$$A + AB = A$$

$$A(1 + B)$$

$$A \cdot 1$$

$$A$$

(8) De Morgan's Law

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The two important laws have been suggested by De Morgan's.

— breaks
x → +
+ → x

(i) This law states that the complement of a product is equal to the addition of the complements.

$$\overline{AB} = \overline{A} + \overline{B}$$

(ii) This law states that the complement of a sum is equal to the product of the complements.

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

Example – 1 : Simplify the expression $A + \bar{A}B + AB$ using Boolean algebra.

$$\begin{aligned} &A + \bar{A}B + AB \\ &A + B(\bar{A} + A) \\ &(A + B) \quad \underline{\text{Ans}} \end{aligned}$$

$$\because \bar{A} + A = 1$$

OR

$$\begin{aligned} &\underline{A + \bar{A}B} + AB \\ &A + B + AB \\ &A + B(1 + A) \\ &(A + B) \quad \underline{\text{Ans}} \end{aligned}$$

$$\begin{aligned} &\because A + \bar{A}B = A + B \\ &1 + A = 1 \end{aligned}$$

Example – 2 : Simplify the Boolean function using Boolean algebra theorems

$$A'B'C' + A'BC' + AB'C' + ABC'$$

$$\underline{\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}}$$

$$\bar{A}\bar{C}(\bar{B} + B) + A\bar{C}(\bar{B} + B)$$

$$\bar{A}\bar{C} + A\bar{C}$$

$$\bar{C}(\bar{A} + A)$$

$$\bar{C}$$

Ans

$$\because \bar{B} + B = 1$$

Example — 3 : Simplify the Boolean function $F = XY + XY'Z + YZ'$ using Boolean algebra.

$$XY + X\bar{Y}Z + Y\bar{Z}$$

$$X(Y + \bar{Y}Z) + Y\bar{Z}$$

$$X(Y + Z) + Y\bar{Z}$$

$$XY + XZ + Y\bar{Z}$$

$$\because Y + \bar{Y}Z = Y + Z$$

It still has 3 terms
ie it is not simplified

$$\therefore XY(1) + XZ + Y\bar{Z}$$

$$XY(Z + \bar{Z}) + XZ + Y\bar{Z}$$

$$XYZ + X\bar{Y}\bar{Z} + XZ + Y\bar{Z}$$

$$XZ(Y + 1) + Y\bar{Z}(X + 1)$$

$$XZ + Y\bar{Z}$$

Ans

Example — 4 : Simplify the following expression $Y = (A+B)(A+C)$.

$$Y = (A + B)(A + C)$$

$$= AA + AC + AB + BC$$

$$= A + AC + AB + BC$$

$$= A(1 + C) + AB + BC$$

$$= A + AB + BC$$

$$= A(1 + B) + BC$$

$$= A + BC$$

Ans

Any logic expression may be expressed in the following two forms

(i) Sum of Product form (SOP)

$$f(A, B, C) = AB + BC + AC$$

Standard or Canonical SOP form

Each product term consists of all the variables in the complemented or un-complemented form.

$$f(A, B, C) = ABC + A\bar{B}\bar{C} + \bar{A}BC$$

(ii) Product of Sum form (POS)

$$f(A, B, C) = (A+B).(B+C).(A+C)$$

Standard or Canonical POS form

Each sum term consists of all the variables in the complemented or un-complemented form.

$$f(A, B, C) = (A + B + C) . (A + \bar{B} + \bar{C}) . (\bar{A} + B + C)$$

Conversion of a logical expression to standard SOP or POS form

Example – 1 : Convert the expression $Y = A B + \bar{A} C + B C$ into the standard SOP form.

$$\begin{aligned}
 Y &= A B + \bar{A} C + B C \\
 &= A B (C + \bar{C}) + \bar{A} C (B + \bar{B}) + B C (A + \bar{A}) \\
 &= A B C + A B \bar{C} + \bar{A} B C + \bar{A} \bar{B} C + A B C + \bar{A} B C \\
 &= A B C + A B \bar{C} + \bar{A} B C + \bar{A} \bar{B} C
 \end{aligned}$$

Example – 2 : Convert the expression $Y = (A+B)(A+C)(B+\bar{C})$ to standard POS form.

$$\begin{aligned}
 Y &= (A+B+C\bar{C})(A+C+B\bar{B})(B+\bar{C}+A\bar{A}) \\
 &= (A+B+C)(A+B+\bar{C})(A+C+B)(A+C+\bar{B})(B+\bar{C}+A)(B+\bar{C}+\bar{A}) \\
 &= (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+\bar{C})
 \end{aligned}$$

Ans.

Minterm

- A product term which contains all the function variables either in complemented form or in un-complemented form is called a Minterm.
- Each individual term in the standard SOP form is called a Minterm

$$Y = AB + \bar{B}C$$

Maxterm

- A sum term which contains all the function variables either in complemented form or in un-complemented form is called a Maxterm.
- Each individual term in the standard POS form is called a Maxterm

$$Y = (A + C) \cdot (B + \bar{C})$$

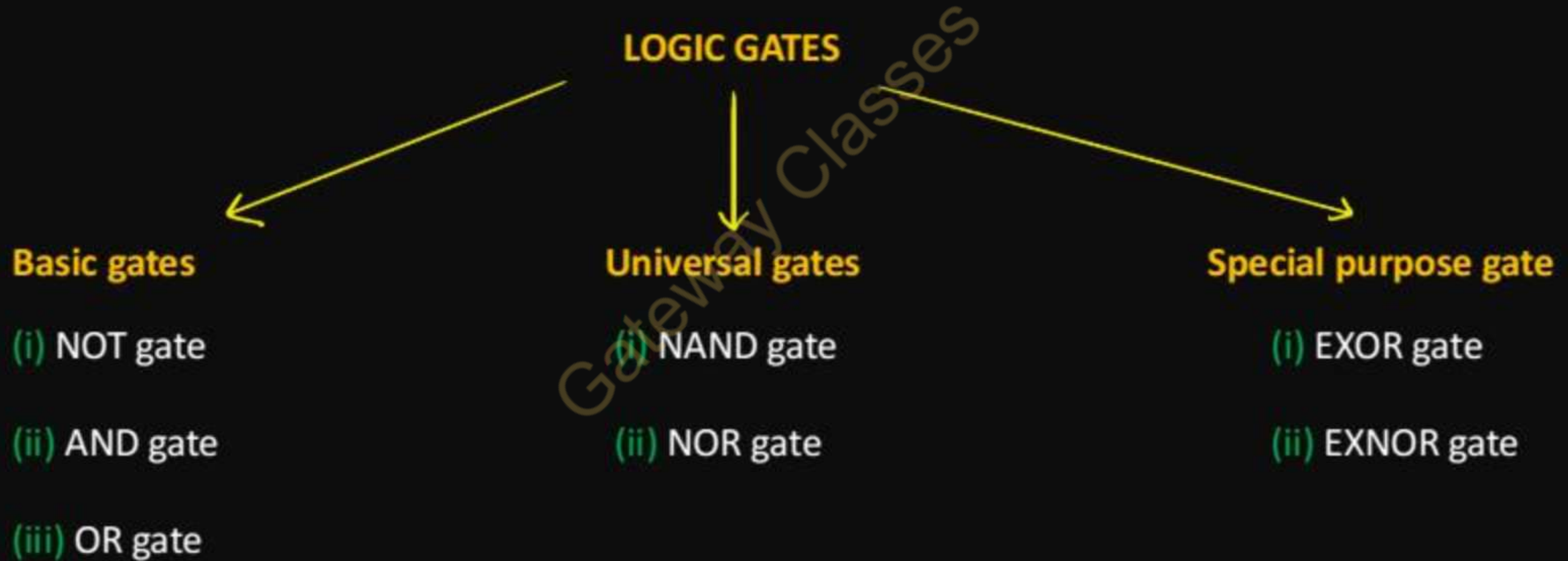
Maximum number of minterms = 2^n

Maximum number of maxterms = 2^n

Where n = number of variables

Variables		Minterms	Maxterms
A	B	m_i	M_i
0	0	$m_0 = \bar{A} \bar{B}$	$M_0 = A + B$
0	1	$m_1 = \bar{A} B$	$M_1 = A + \bar{B}$
1	0	$m_2 = A \bar{B}$	$M_2 = \bar{A} + B$
1	1	$m_3 = AB$	$M_3 = \bar{A} + \bar{B}$

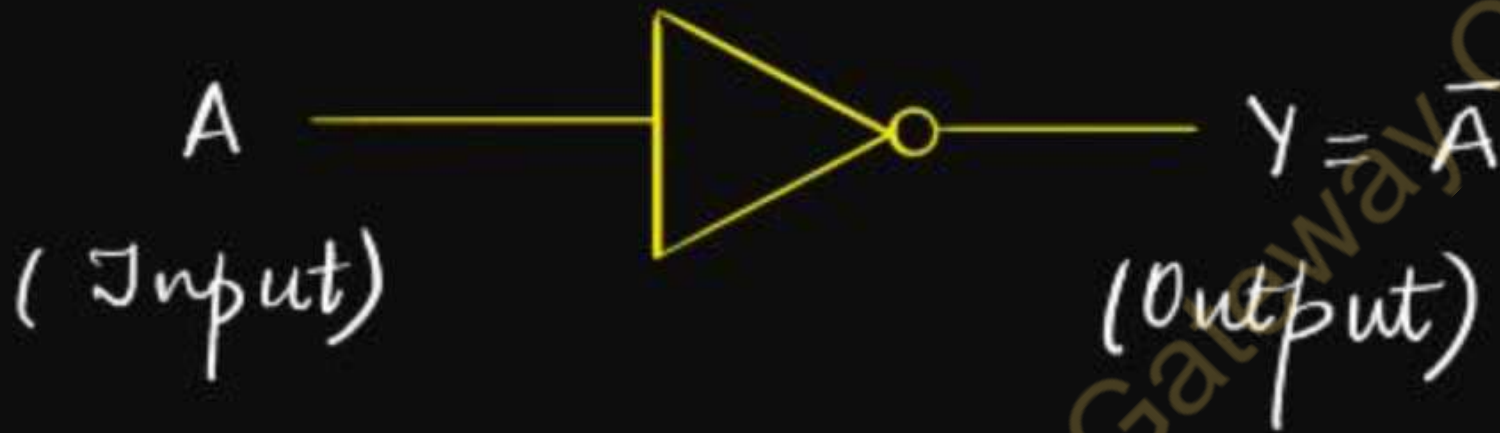
- The most basic element of digital circuits are the logic gates.
- Boolean functions may be practically implemented by using Logic gates.
- Electronic gates require a power supply.



(1) NOT Gate

- The NOT gate is also known as an inverter.
- Output is the inverted version or complement of its input.
- NOT gate has only one input and one output.

Symbol



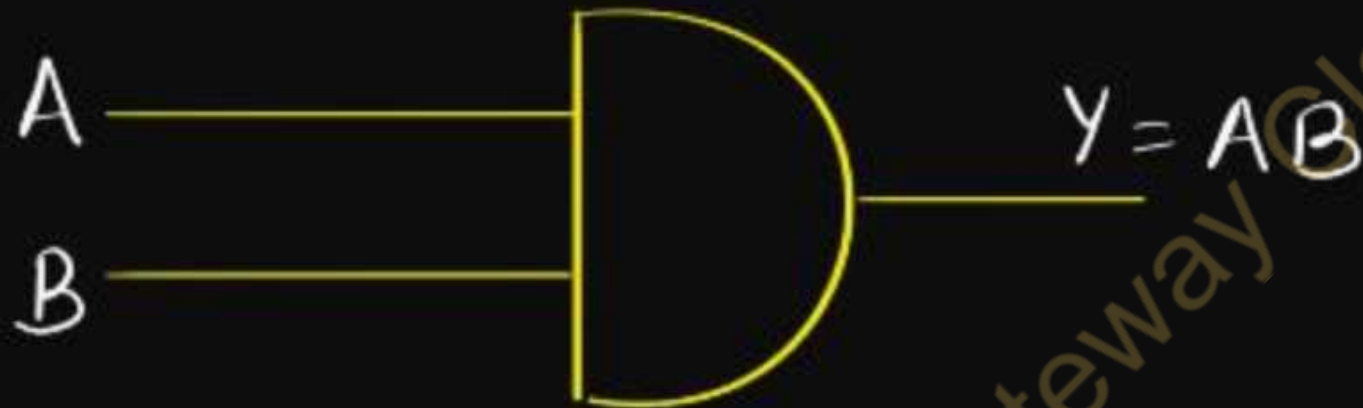
Truth Table

Input	Output
A	$Y = \bar{A}$
0	1
1	0

(2) AND gate

- The AND gate is an electronic circuit that gives a high output (1) only if all its inputs are high.
- It performs logical multiplication on its inputs.

Symbol



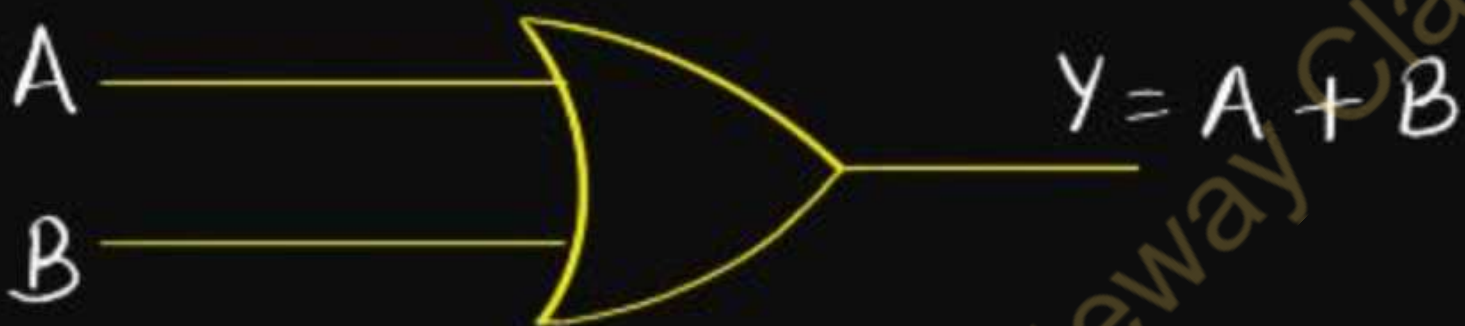
Truth Table

Input		Output
A	B	$Y = A . B$
0	0	0
0	1	0
1	0	0
1	1	1

(3) OR gate

- The OR gate is an electronic circuit that gives a high output (1) if one or more of its inputs are high.
- It performs logical addition on its inputs.

Symbol



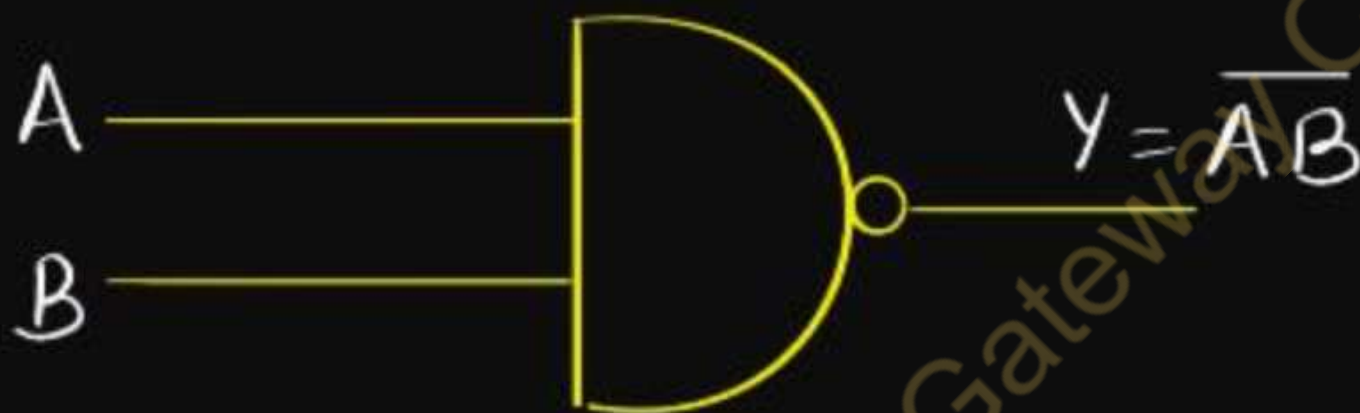
Truth Table

Input		Output
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

(4) NAND gate

- NOT + AND gate
- The input of all NAND gates are high if any of the input are low
- The small circle represents inversion

Symbol



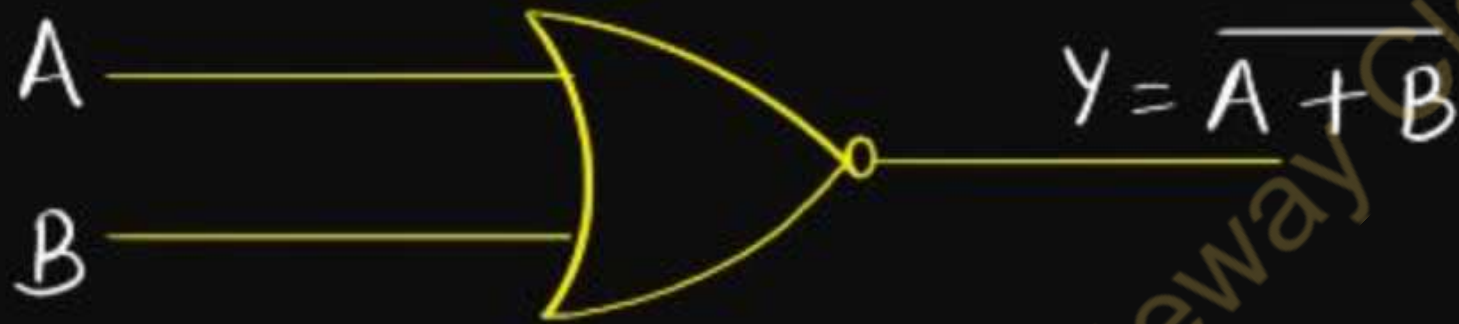
Truth Table

Input		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

(5) NOR gate

- Not + OR Gate
- The outputs of all NOR gates are low if any of the inputs are high.

Symbol



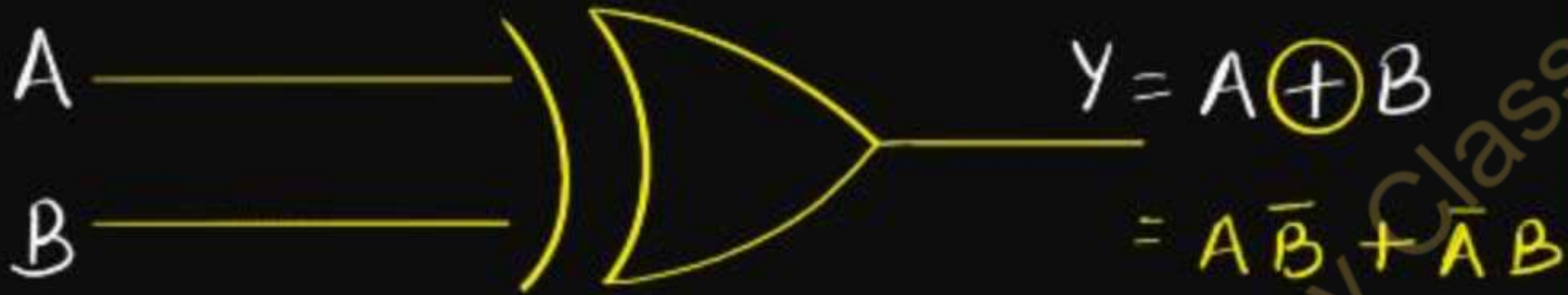
Truth Table

Input		Output
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

(6) EXOR gate

- The EXOR gate is a circuit which will give a high output if both the input are different
- $\oplus$ is used to show EXOR operation

Symbol



A	B	$\bar{A}$	$\bar{B}$	$A\bar{B}$	$\bar{A}B$	$Y = A\bar{B} + \bar{A}B$
0	0	1	1	0	0	0
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	0	0	0

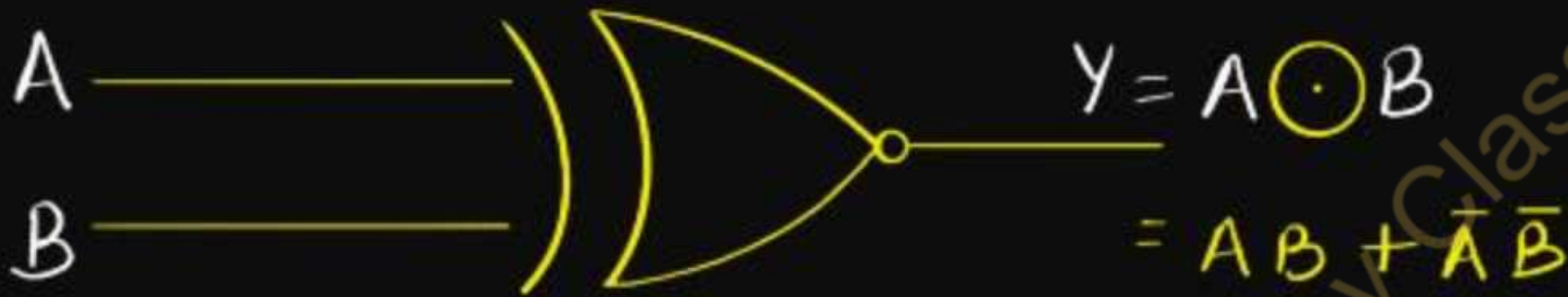
Truth Table

Input		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

(7) EX – NOR gate

- The EX-NOR gate is a circuit which will give a high output if both the input are same.
- $\odot$ is used to show EX-NOR gate.

Symbol



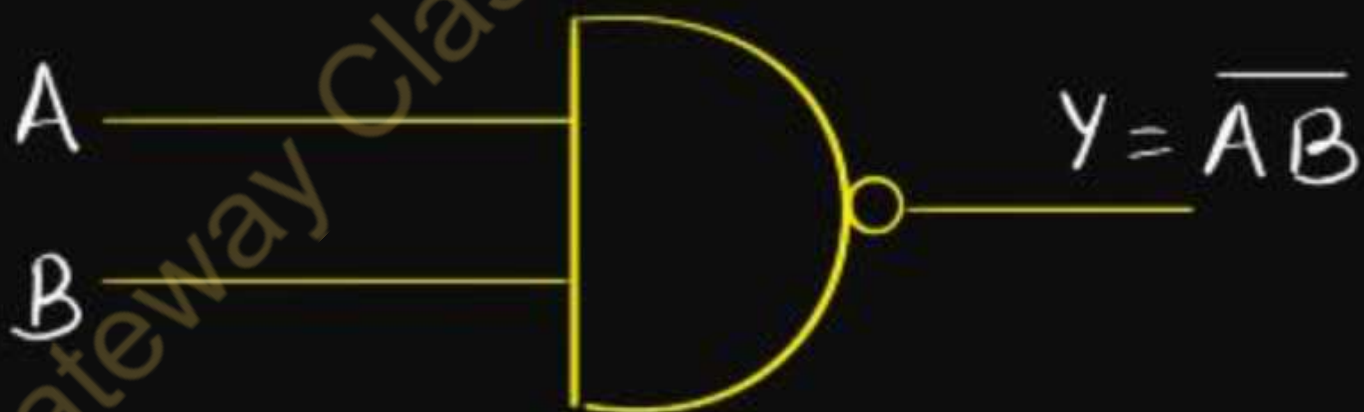
A	B	$\bar{A}$	$\bar{B}$	AB	$\bar{A}\bar{B}$	$Y = AB + \bar{A}\bar{B}$
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1

Truth Table

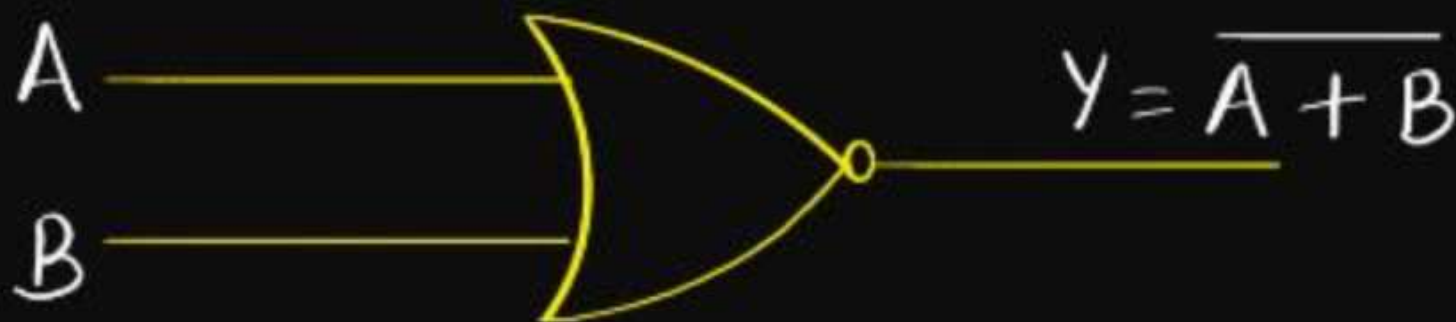
Input		Output
A	B	$Y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

- NAND and NOR gates are called as universal gates because it is possible to implement any Boolean expression with the help of only NAND gates or only NOR gates.
- Any logical gates or combinational circuit can be build with the help of only NAND gates or only NOR gates.

NAND Gate

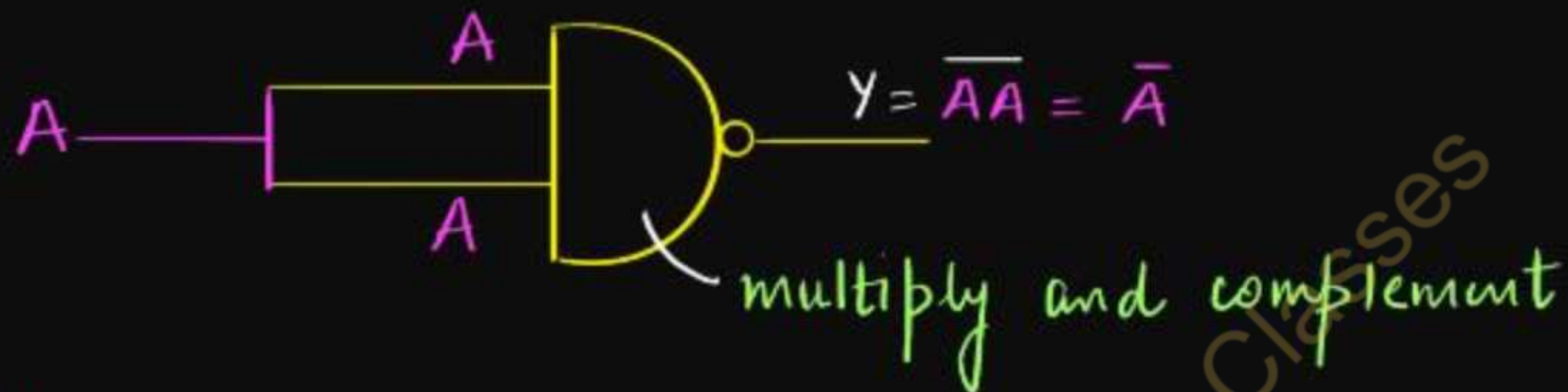


NOR Gate

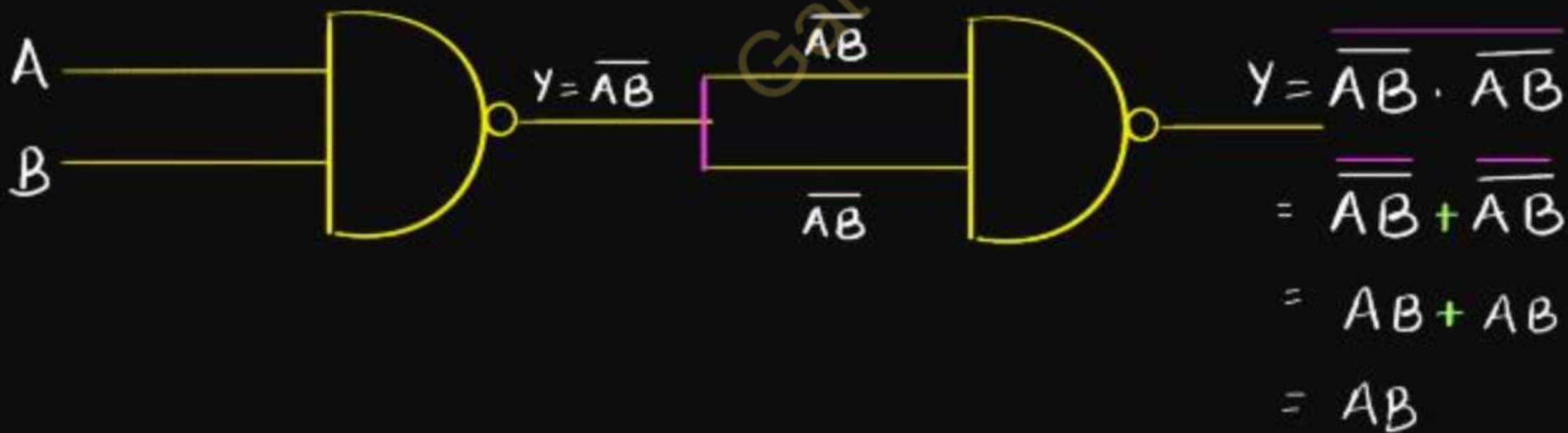


NAND Gate as a Universal Gate

(1) NOT gate using NAND gate

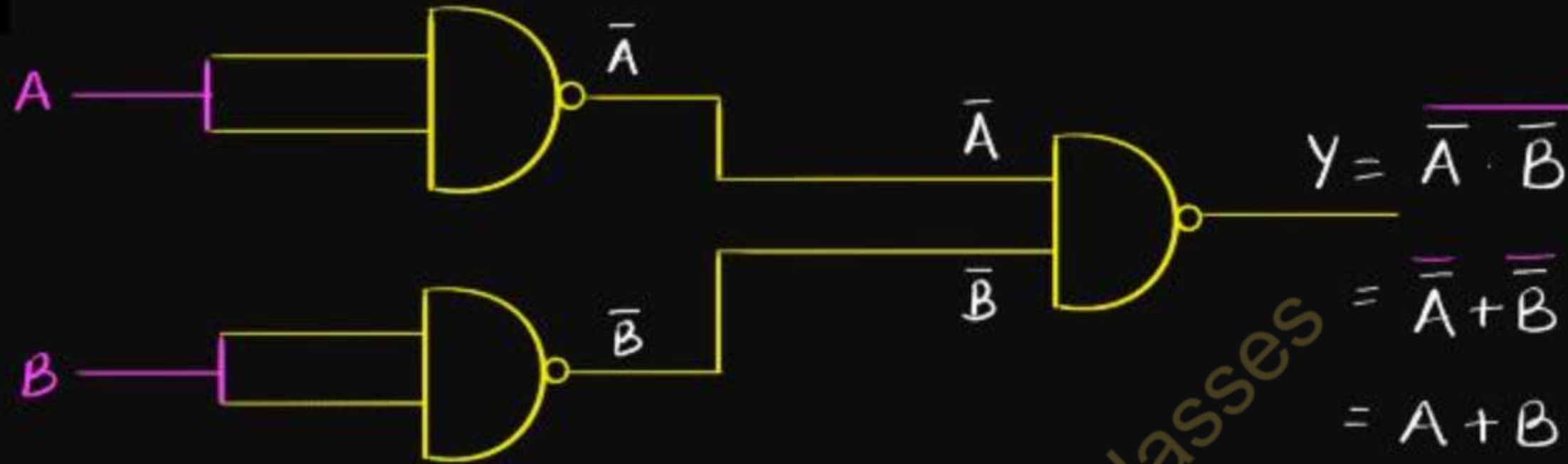


(2) AND gate using NAND gate

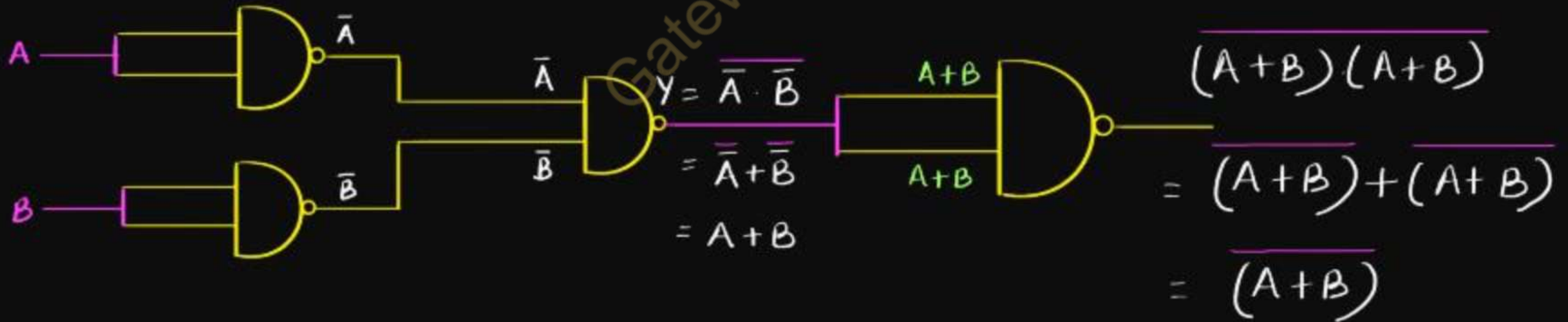


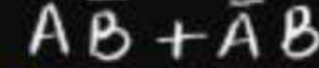
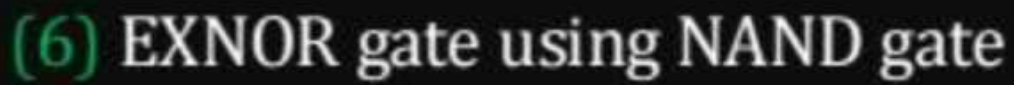
	1
	2
start from A & B,	3
"	4
"	5 EXOR
"	6 EXNOR

(3) OR gate using NAND gate



(4) NOR gate using NAND gate

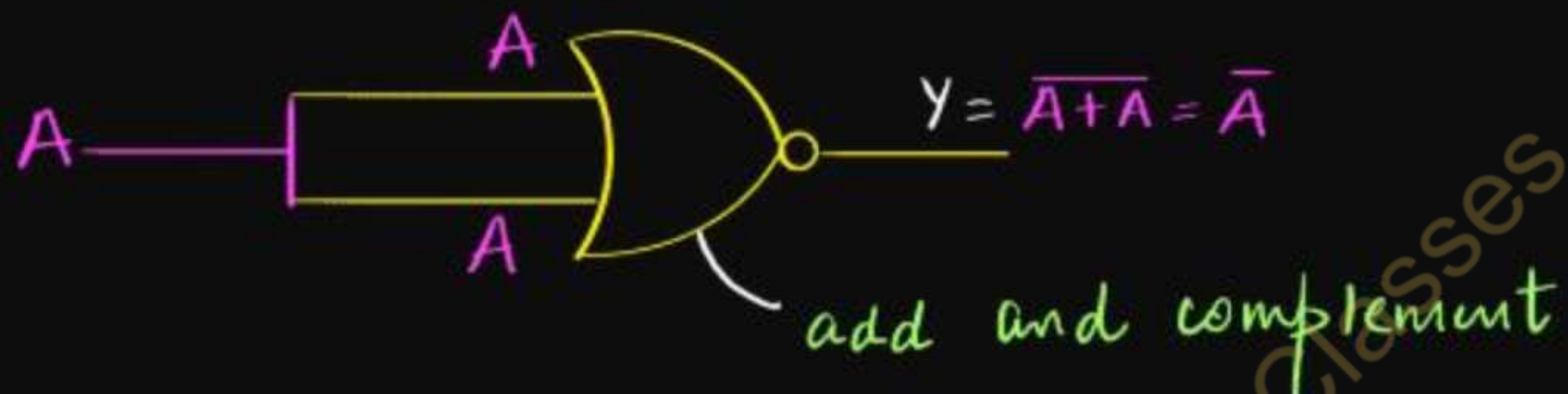




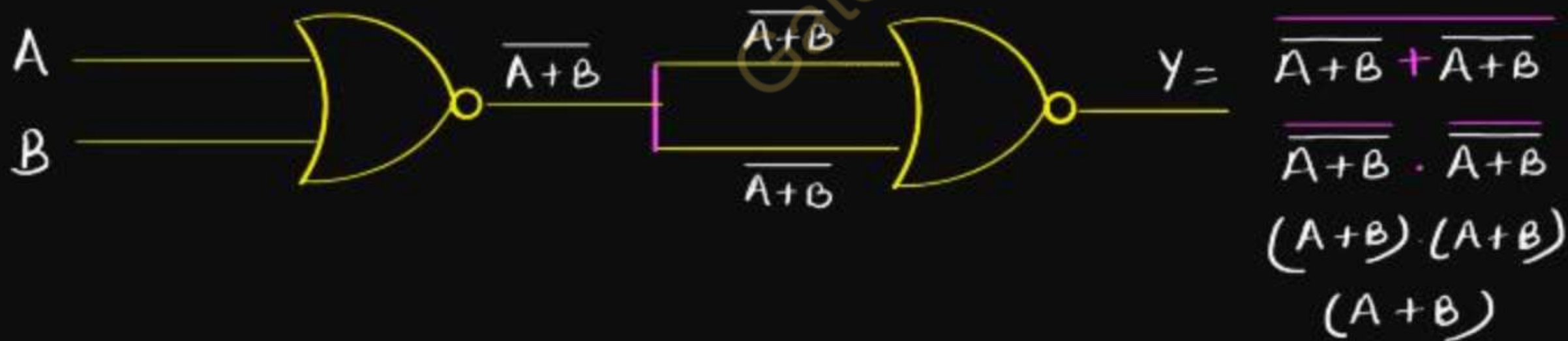
$$= A \odot B$$

NOR gate as a universal gate

(1) NOT gate using NOR gate

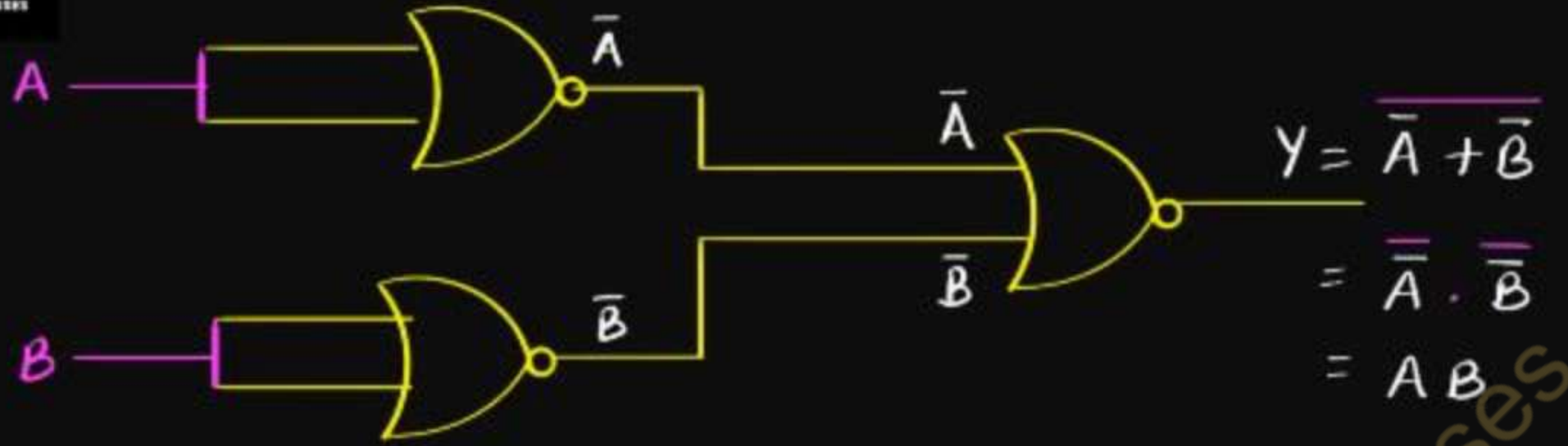


(2) OR gate using NOR gate

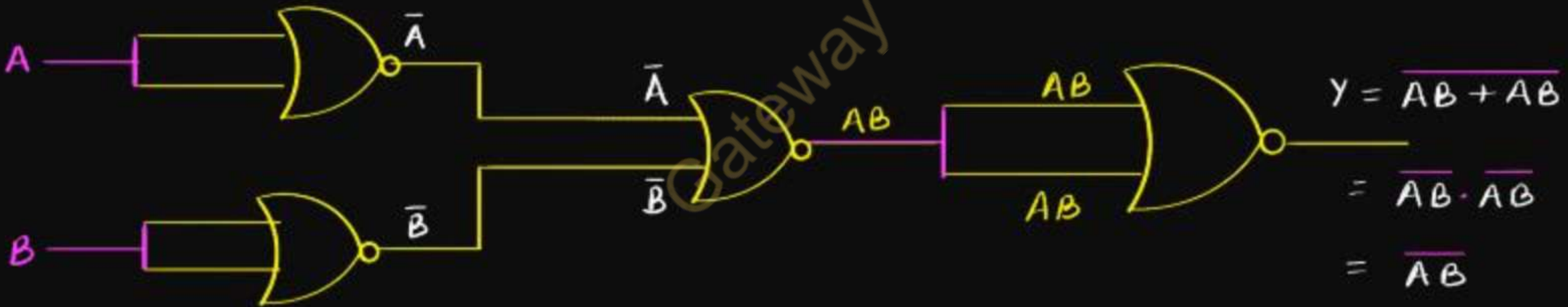


	1
	2
start from A & B,	3
"	4
"	5 EX NOR
"	6 EX OR

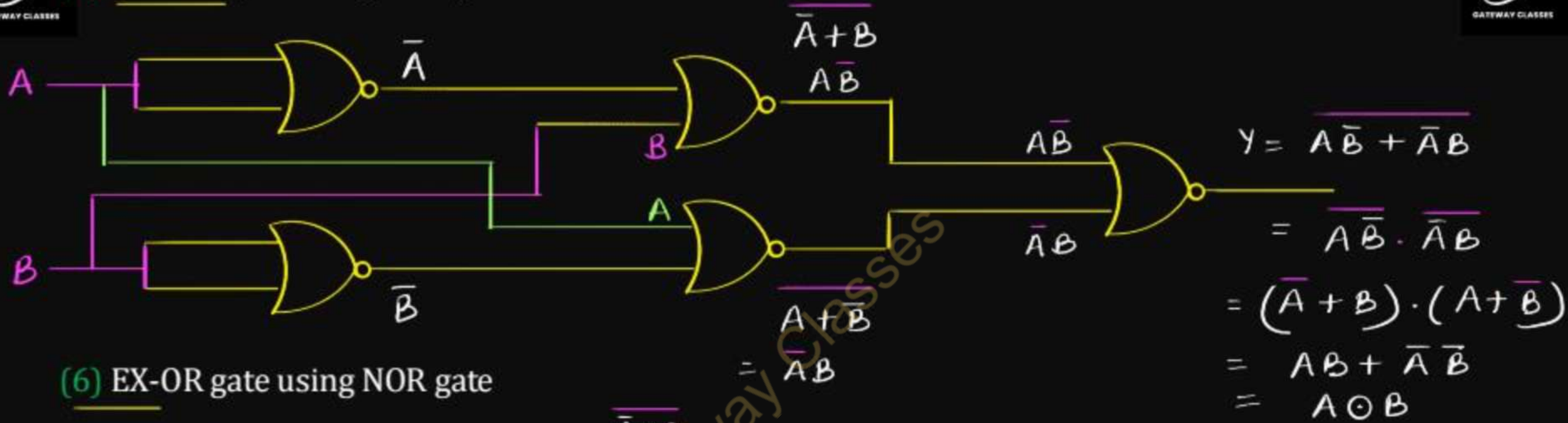
(3) AND gate using NOR gate



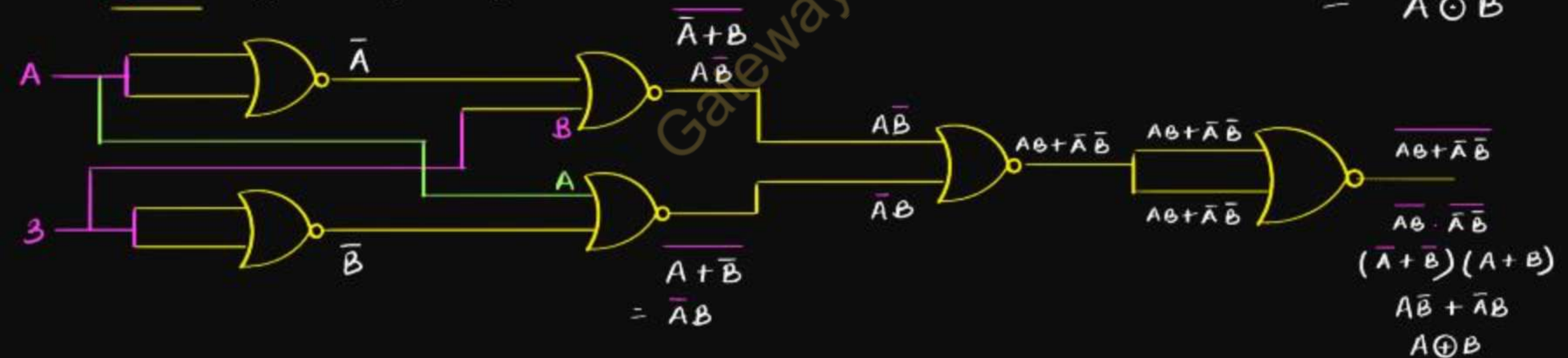
(4) NAND gate using NOR gate



(5) EX-NOR gate using NOR gate



(6) EX-OR gate using NOR gate



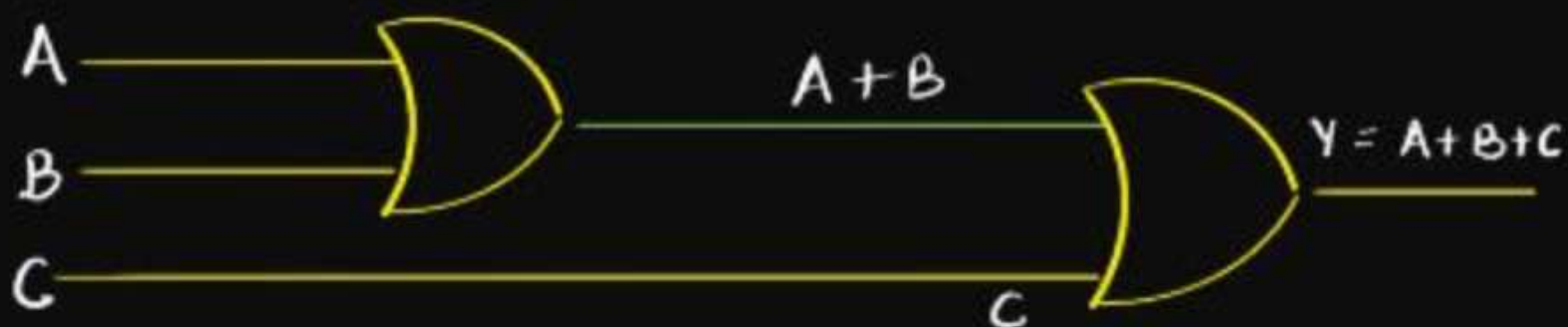
Que) Realize a 3-input gate using 2-input gates for the following gates:

(i) AND (ii) OR (iii) NAND (iv) NOR

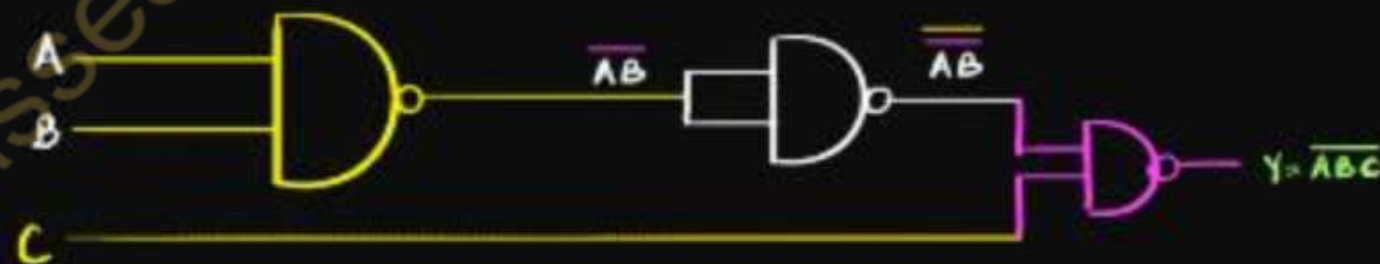
(i)



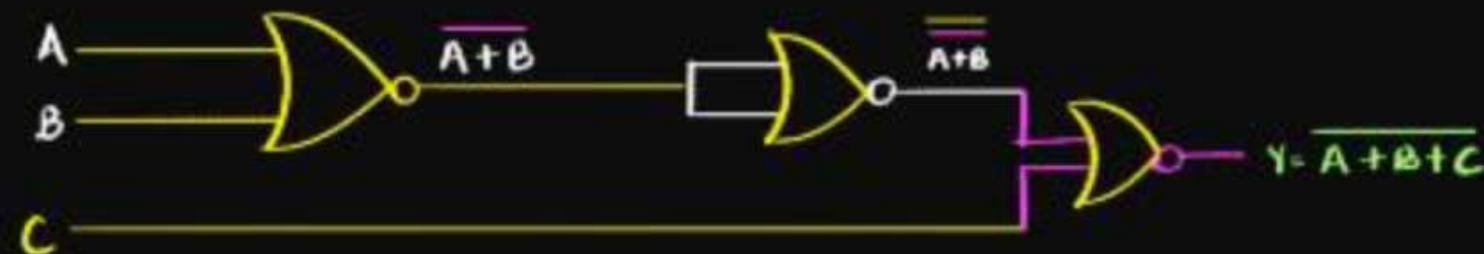
(ii)



(iii)



(iv)



Karnaugh Map (K-map)

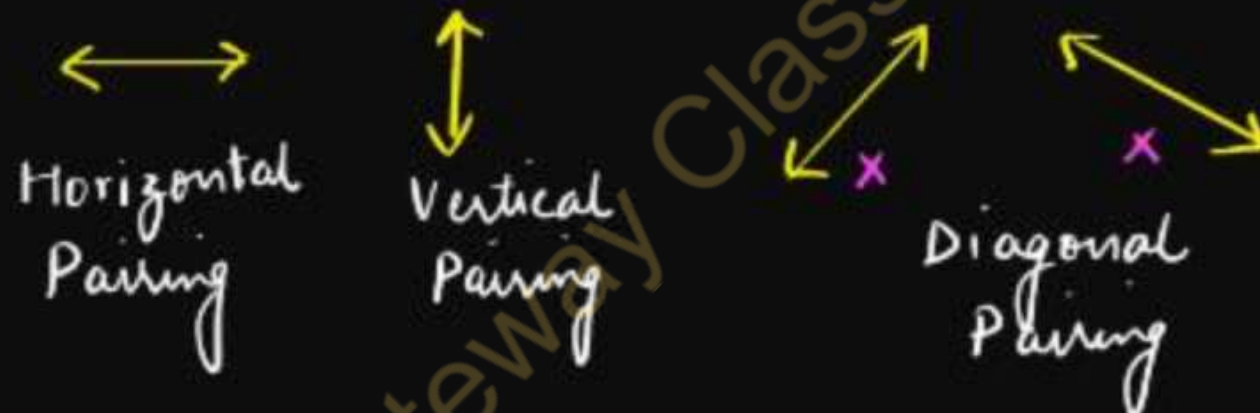
- The K-map is a pictorial representation of truth table of the Boolean function and used to minimize the Boolean function
- K-map matrix contain 2^n cell, where n is the total number of variables
- > K-maps are of 2, 3, 4, 5, 6 variables
- > Disadvantage (i) Cannot be solved using computers
(ii) Are complex when number of variables > 5 .



Rule of Minimization

- **Formation of Groups** : Form minimum number of groups and Form big size groups
- K maps are Foldable
- Check for Redundant Groups, if any don't make them.
- Focus on pairing of 1's & 0's using don't care if given but do not focus on pairing of Don't cares.

Rules of Pairing



- All 4 corners can be paired in a Quad (Group of Four)
- Grouping is allowed only in 5-variable, 6-variable K-Map
not in 3-variable and 4-variable K-Map.



K-Map for SOP Form

2 Variable K-Map

$2^2 = 4 \rightarrow$ 4 entries in Truth Table

4 cells in K-Map (Cell no. 0 to (n-1) i.e (0 to 3))

- Steps
- 1) Filling
 - 2) Pairing
 - 3) Extraction of ans.

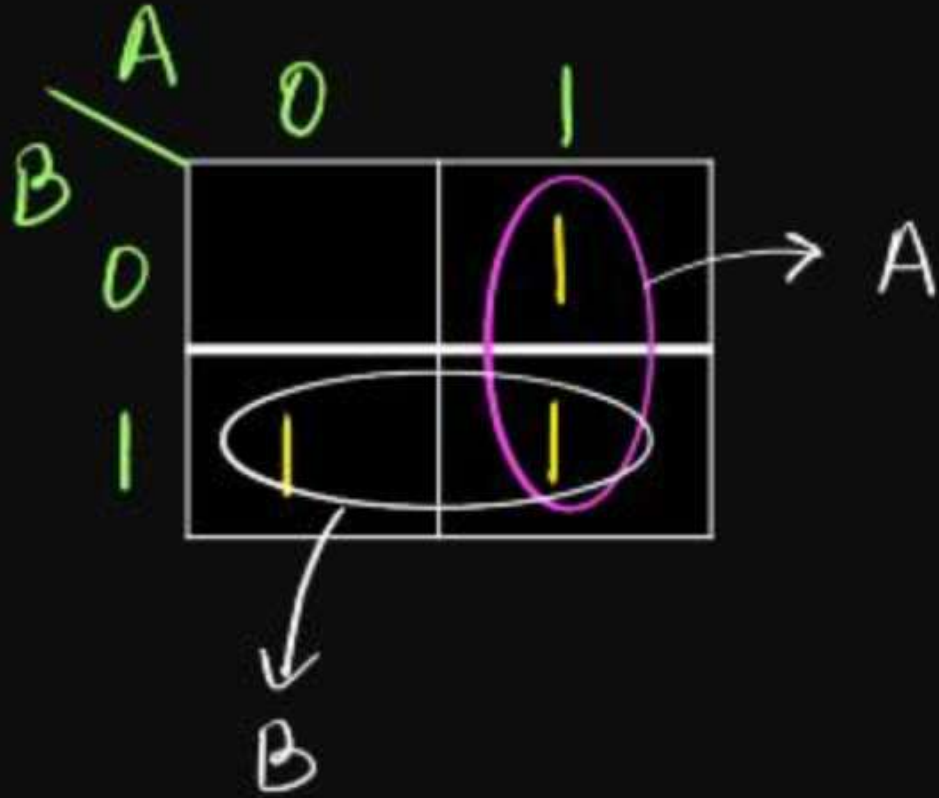
A	B	Cell No.
0	0	0
0	1	1
1	0	2
1	1	3

		A	
		0	1
B	0		
	1		

		A	
		0	1
B	0	0	2
	1	1	3

Que) Minimize using K-map

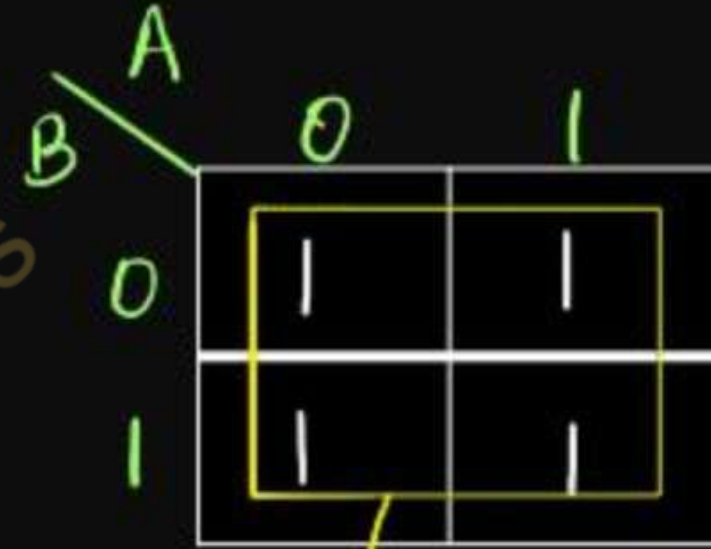
$$Y = AB' + A'B + AB$$



$$Y = A + B$$

Que) Simplify the following function using K-map

$$F(A, B) = \sum m(0, 1, 2, 3)$$



Nothing is common amongst all four 1's
∴ ans is 1

$$F(A, B) = 1$$

- Steps
- 1) Filling
 - 2) Pairing
 - 3) Extraction of ans

3 Variable K-Map

$2^3 = 8 \rightarrow$ 8 entries in Truth Table

8 cells in K-Map (Cell no. 0 to (n-1) i.e (0 to 7))

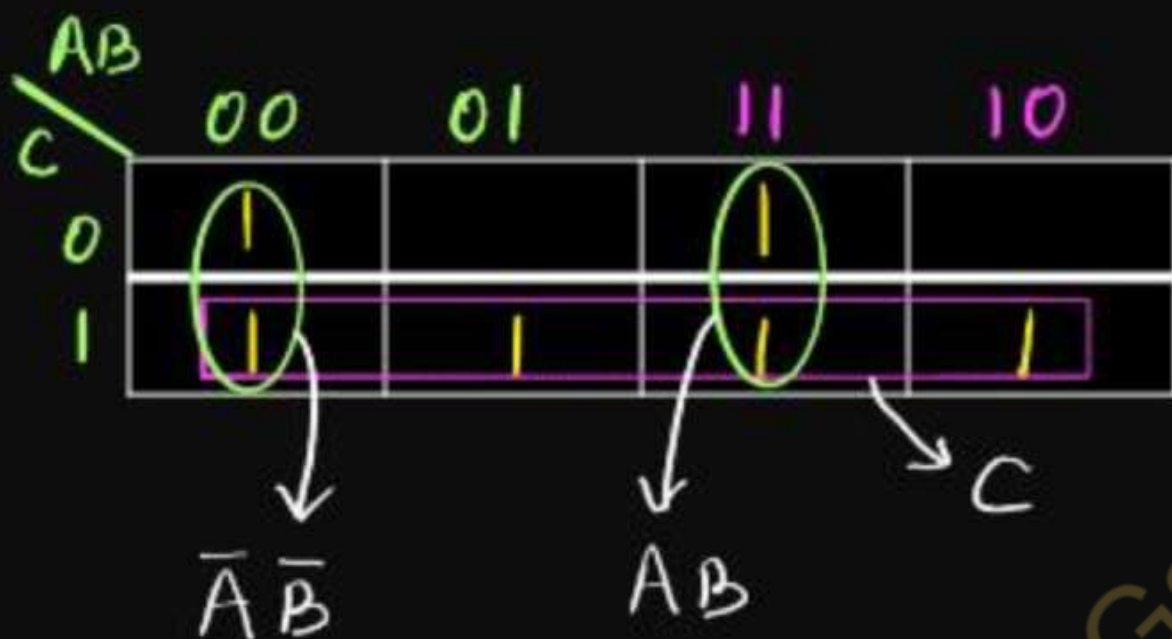
A	B	C	Cell No.
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

	AB	00	01	11	10
C	0				
	1				

	AB	00	01	11	10
C	0	0	2	6	4
	1	1	3	7	5

Que) Simplify the following
function using K-map

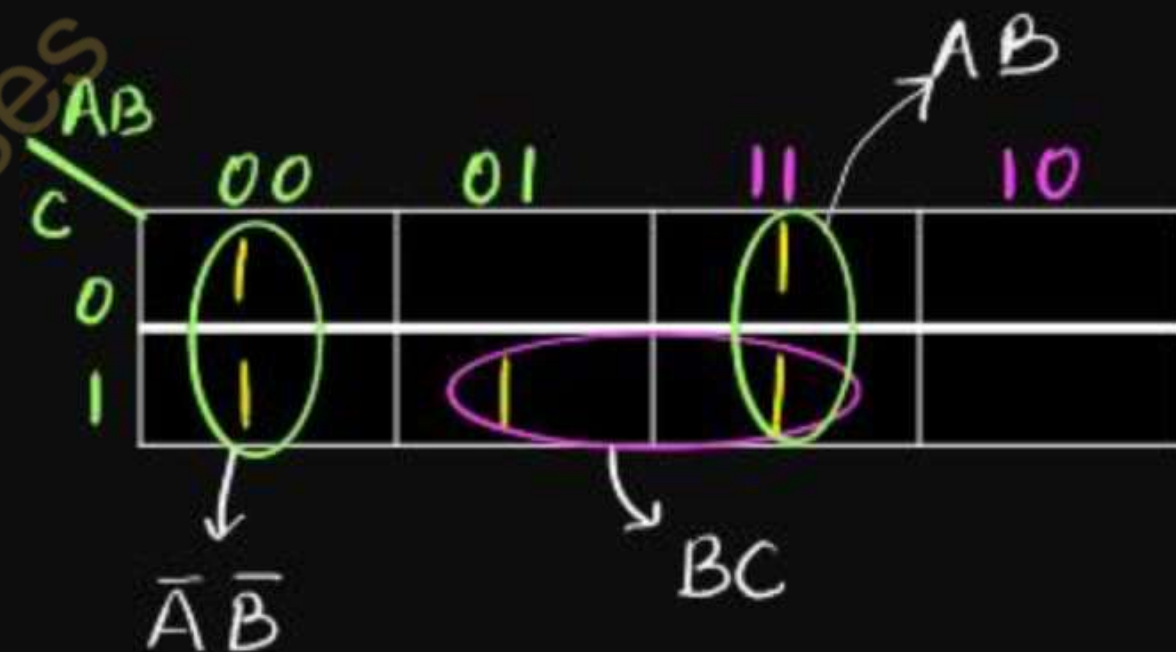
$$F(A, B, C) = \sum m(0, 1, 3, 5, 6, 7)$$



$$F(A, B, C) = AB + \bar{A}\bar{B} + C$$

Que) Simplify the following
function using K-map

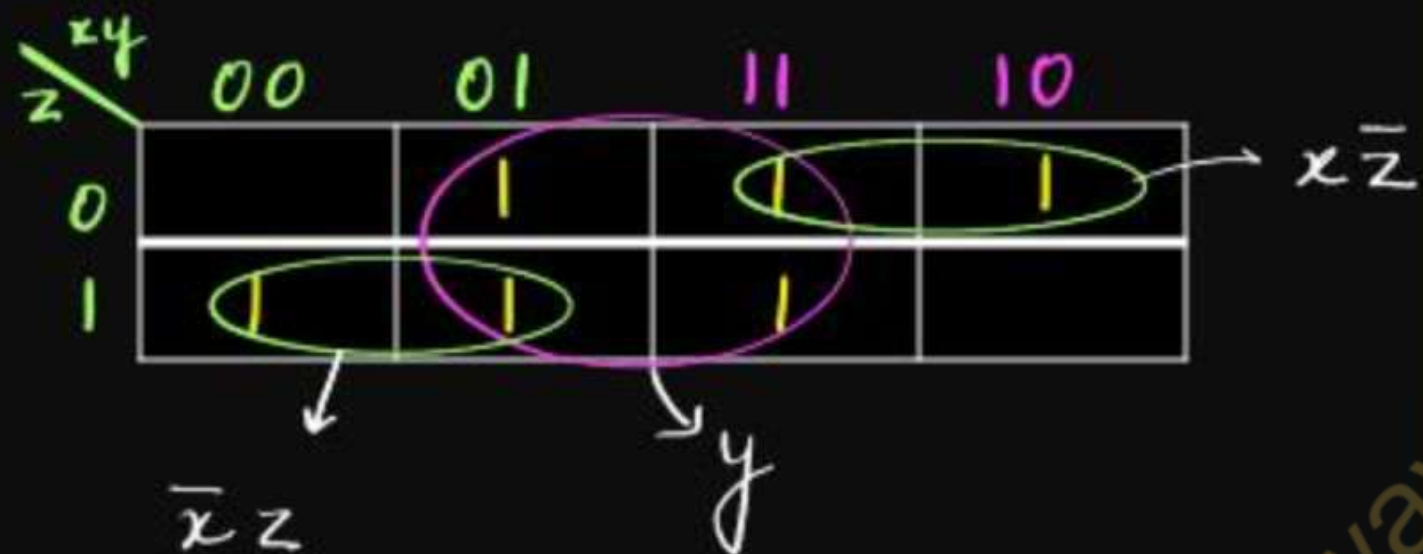
$$F(A, B, C) = \sum m(0, 1, 3, 6, 7)$$



$$F(A, B, C) = AB + \bar{A}\bar{B} + BC$$

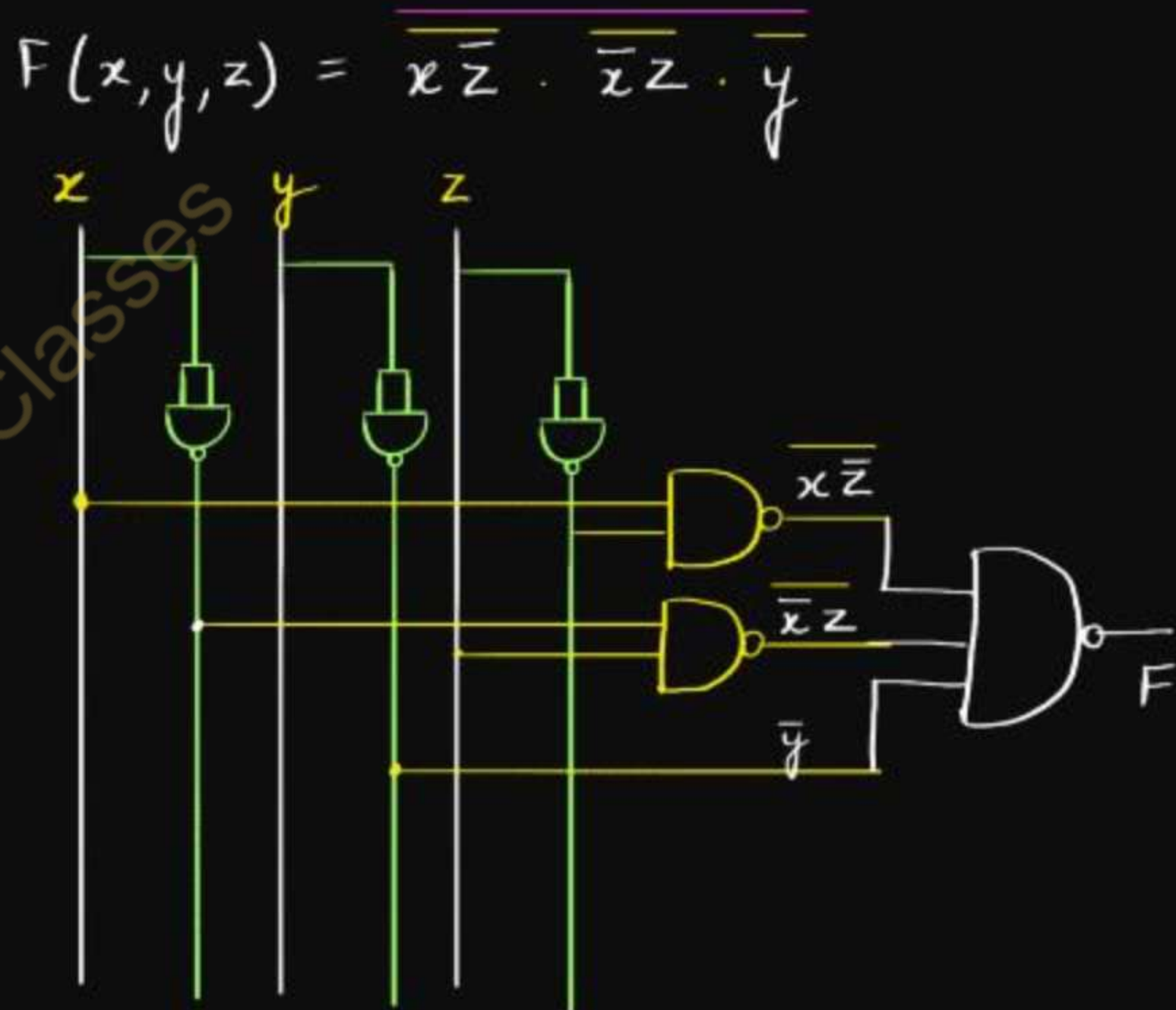
Que) Implement the Boolean function $F(x,y,z) = \sum m(1,2,3,4,6,7)$ using NAND gates.

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$F(x,y,z) = xz̄ + x̄z + y$ Ans
 Implementation Using Nand Gates only.

$F(x,y,z) = \overline{\overline{xz̄ + x̄z + y}}$



A	B	C	D	Cell No.
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

4 Variable K-Map

$2^4 = 16 \rightarrow$ 16 entries in Truth Table

16 cells in K-Map (Cell no. 0 to (n-1) i.e (0 to 15))

		AB			
		00	01	11	10
CD	00	0	4	12	8
	01	1	5	13	9
	11	3	7	15	11
	10	2	6	14	10

Que) Minimize the following Boolean function using K-Map

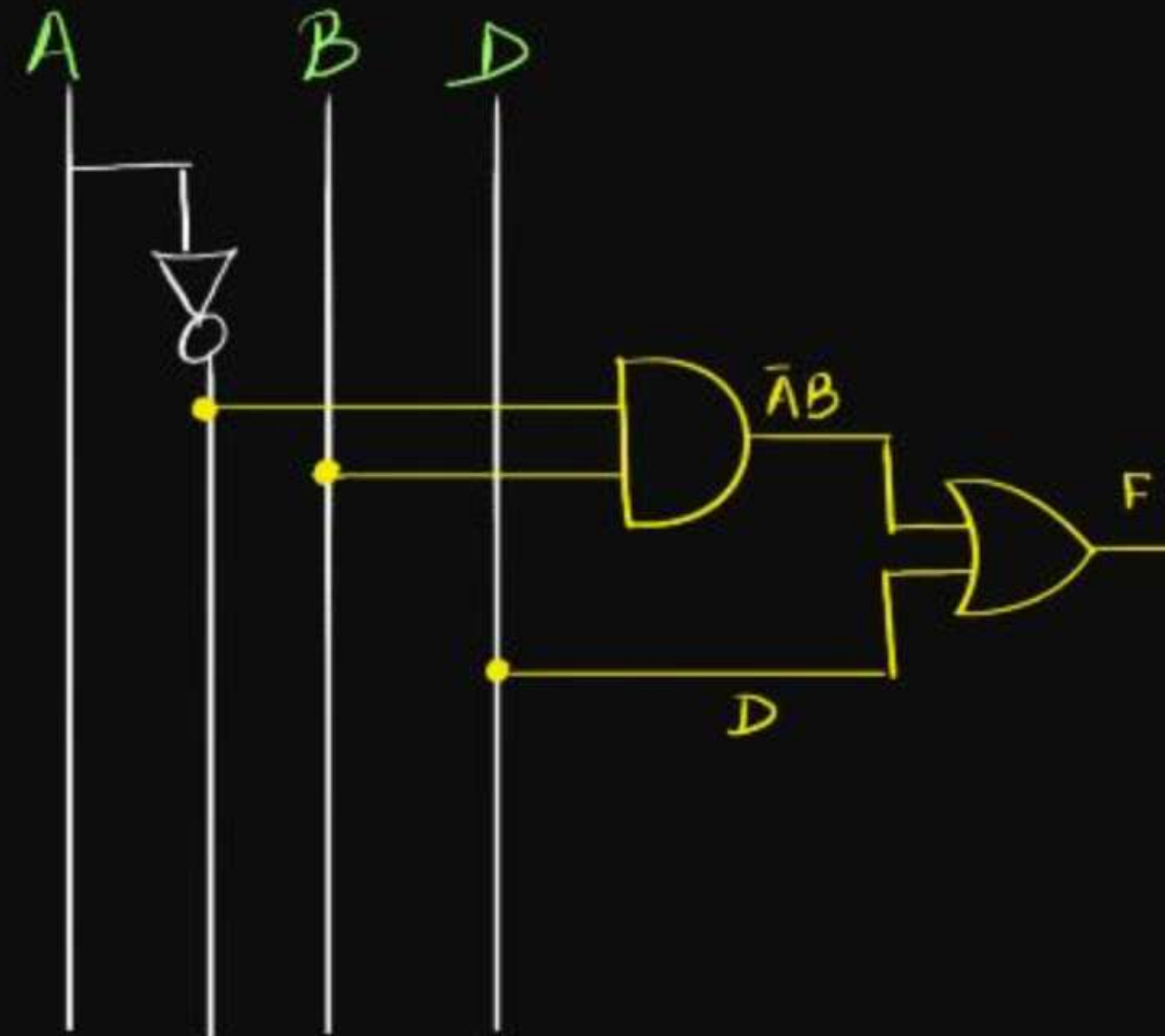
$$F(A, B, C, D) = \Sigma (1, 3, 4, 5, 6, 7, 9, 11, 13, 15).$$

Also implement the simplified function using basic gates only.

AB \ CD	00	01	11	10
00		1		
01	1	1	1	1
11	1	1	1	1
10		1		

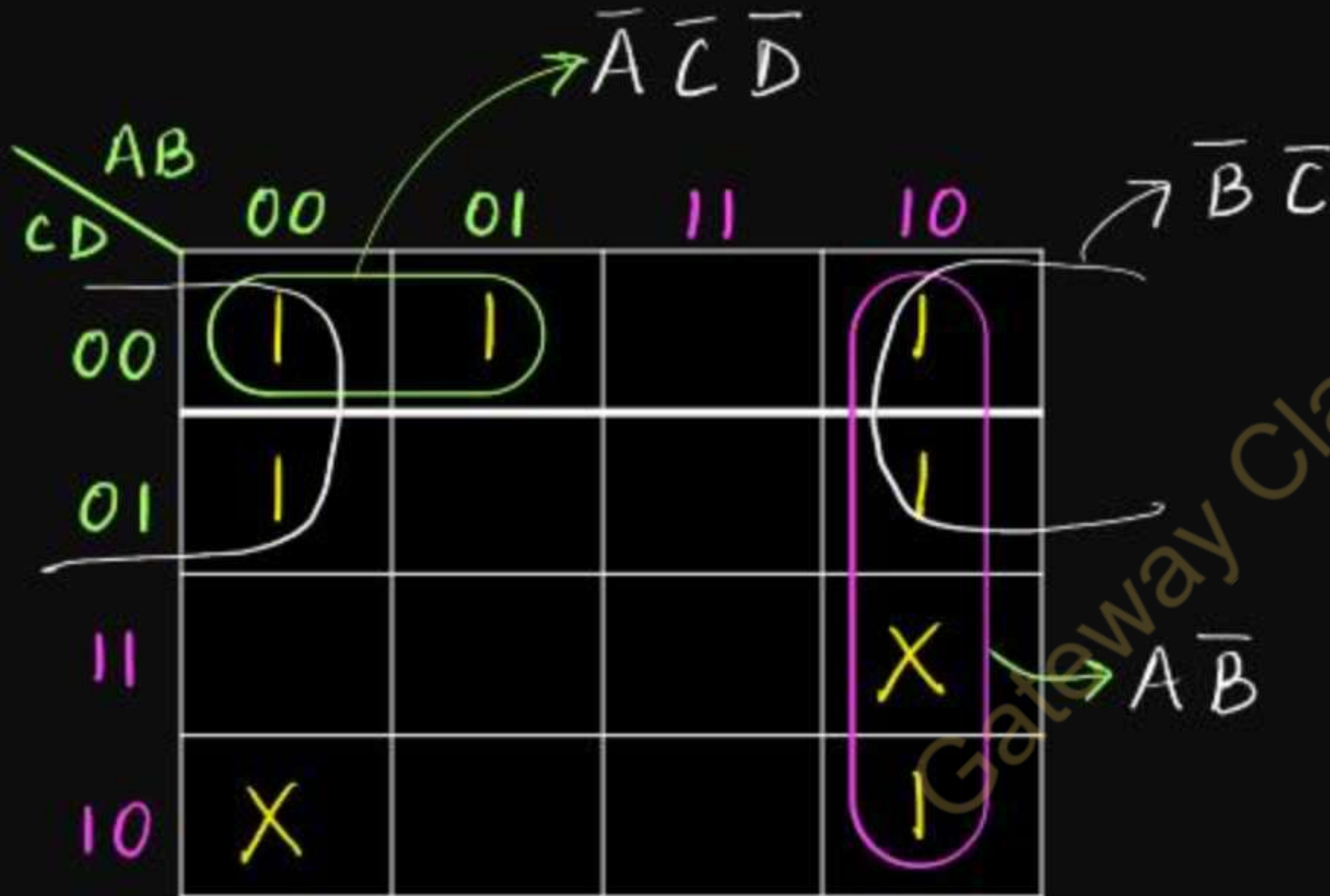
$\bar{A}B$

$$F(A, B, C, D) = \bar{A}B + D$$



Que) Minimize the following Boolean function using K-Map
 $F(A,B,C,D) = \sum m (0,1,4,8,9,10) + \sum d (2,11)$

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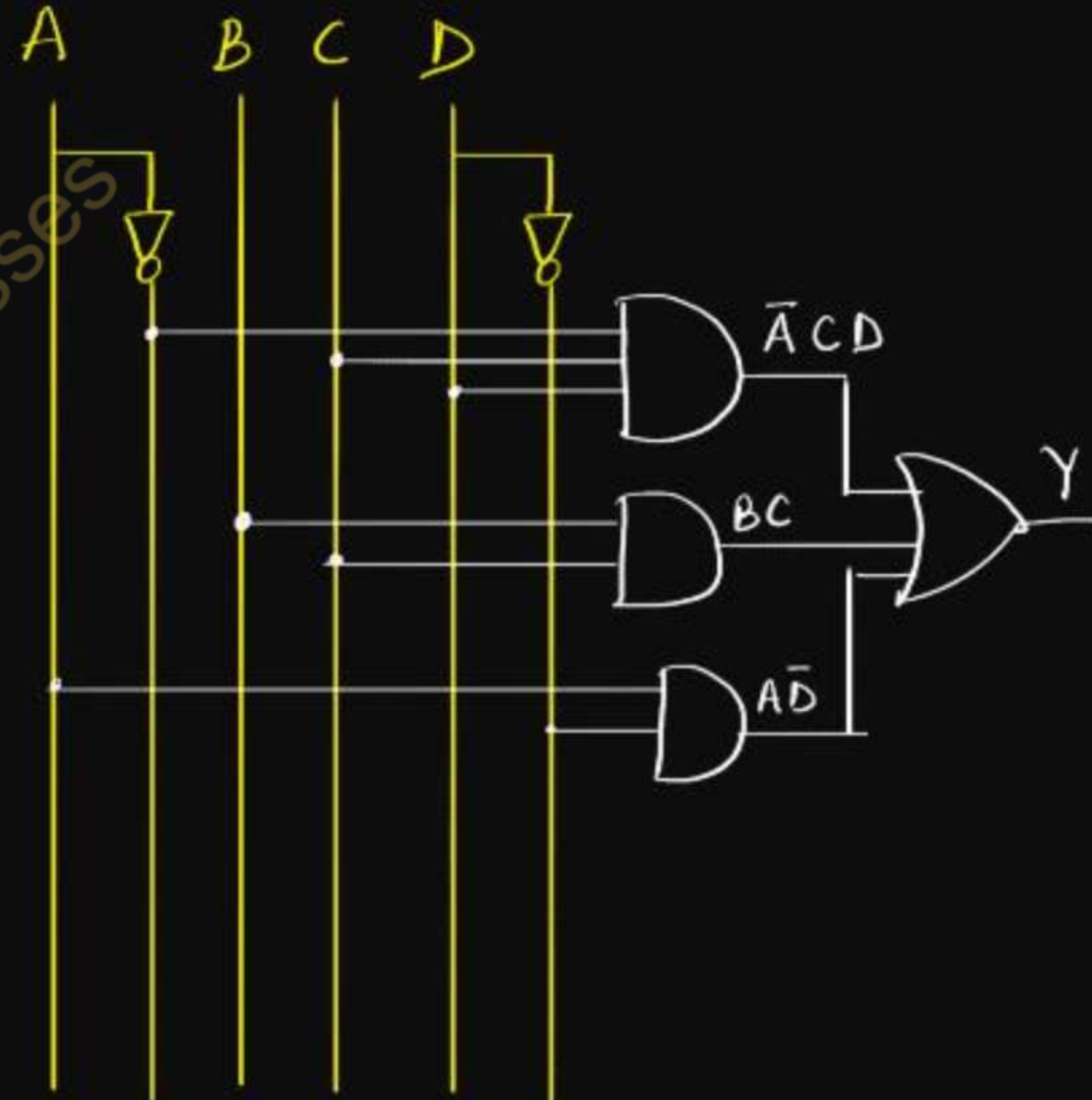
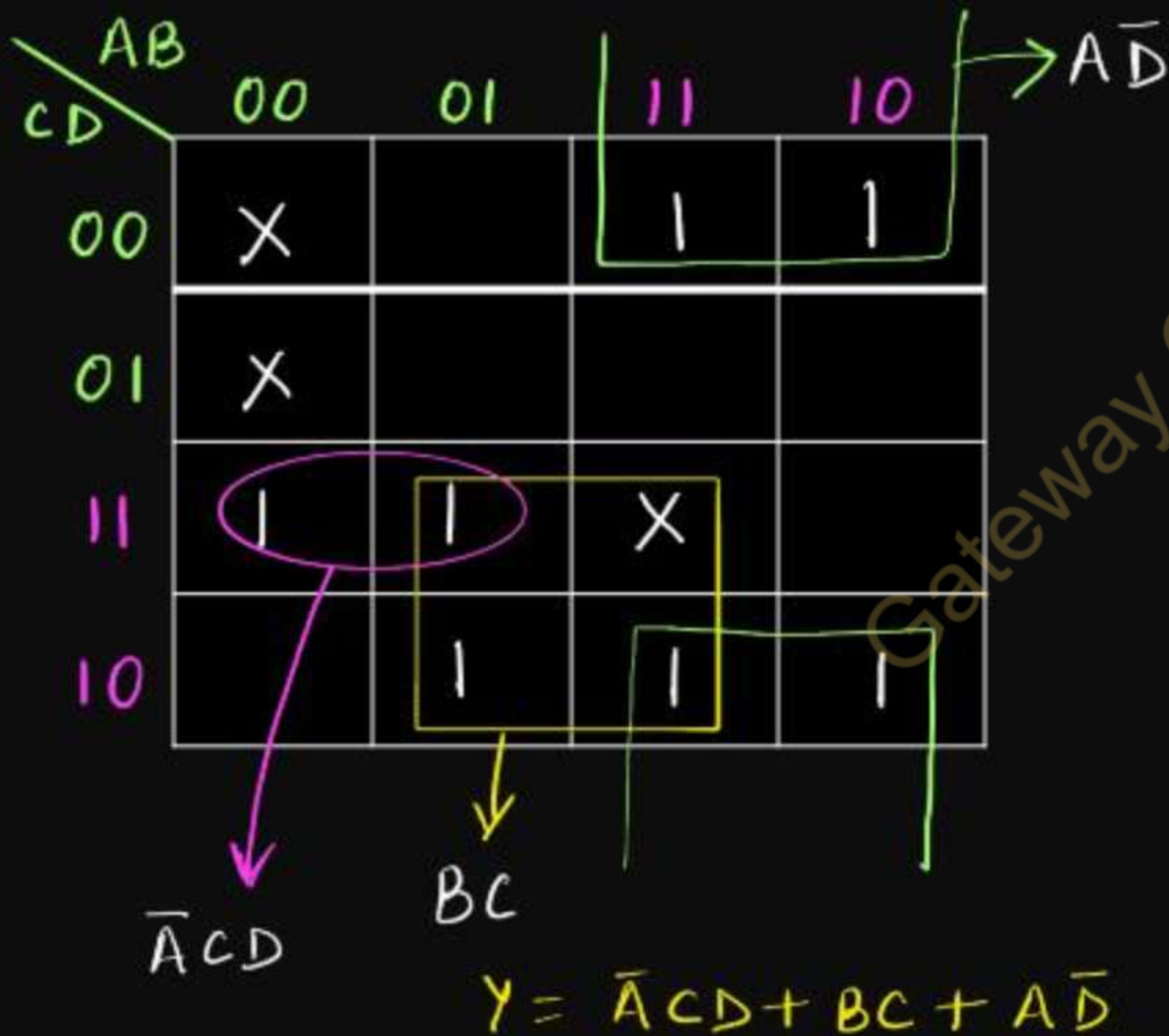


$$F(A,B,C,D) = A\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}\bar{D}$$

No need to make
all 4 corners.

Que) Simplify $Y = \sum m (3, 6, 7, 8, 10, 12, 14) + d (0, 1, 15)$ using K-map method and implement the simplified circuit using logic gates.

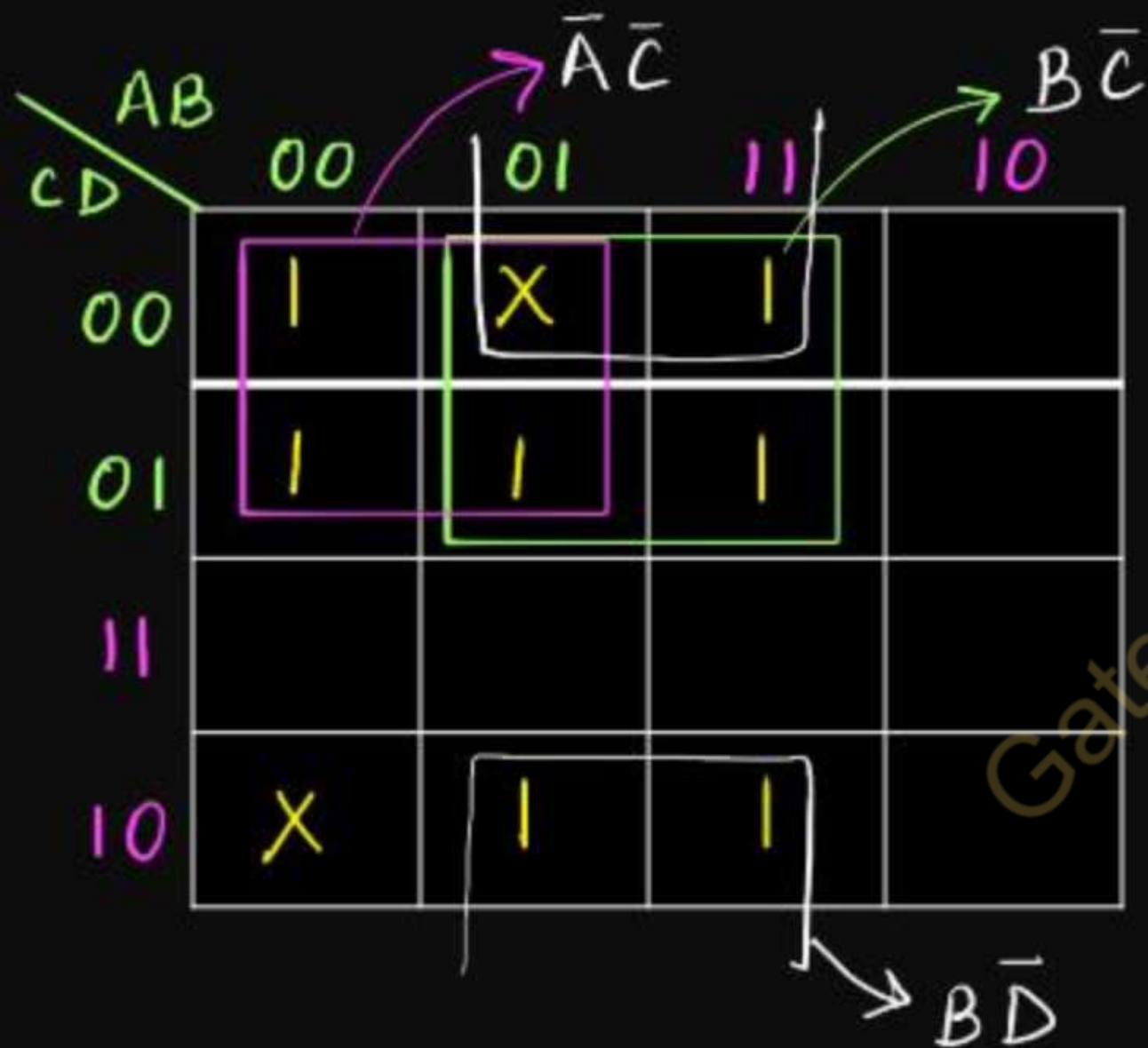
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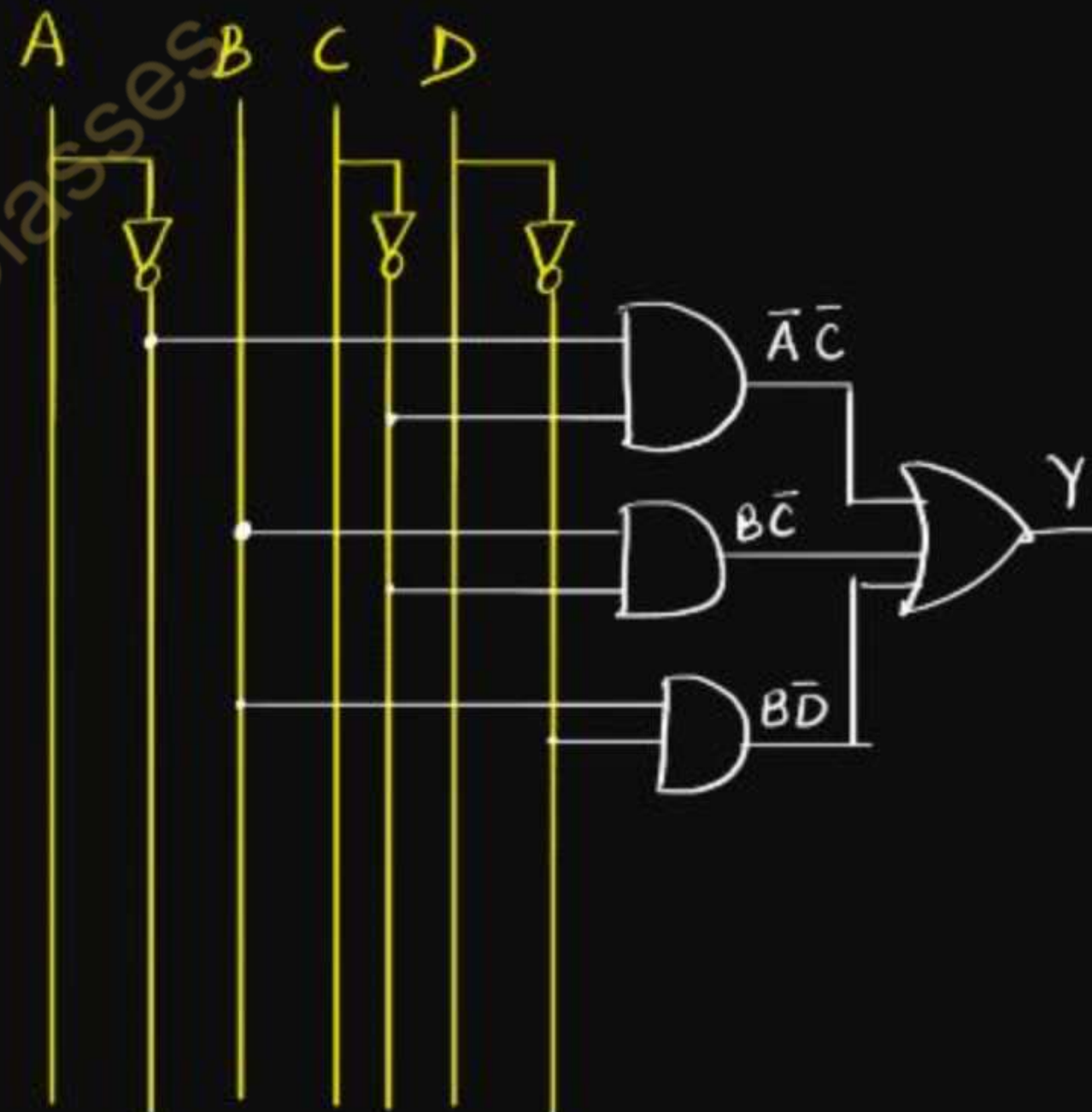
Que) Discuss the Karnaugh map (K-map) method for Boolean function using K-Map and also draw the simplified logic circuit using basic gates.

$$F(A,B,C,D) = \sum m (0,1,5,6,12,13,14) + d (2,4)$$

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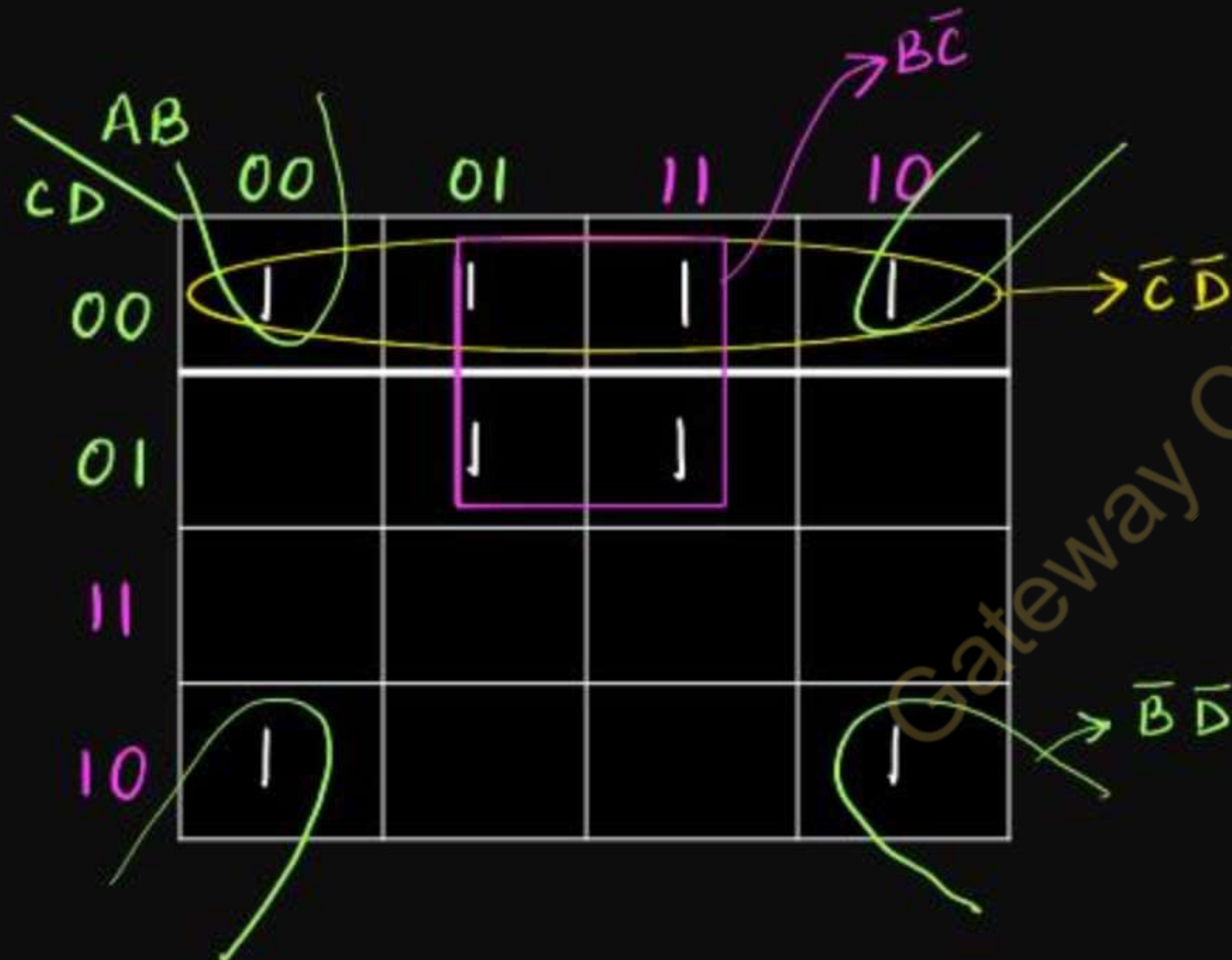
$$F = \bar{A}\bar{C} + B\bar{C} + B\bar{D}$$



Que) Using K-map, minimize the Boolean function and implement it using only NAND gates

$$F(A,B,C,D) = \sum m(0,2,4,5,8,10,12,13).$$

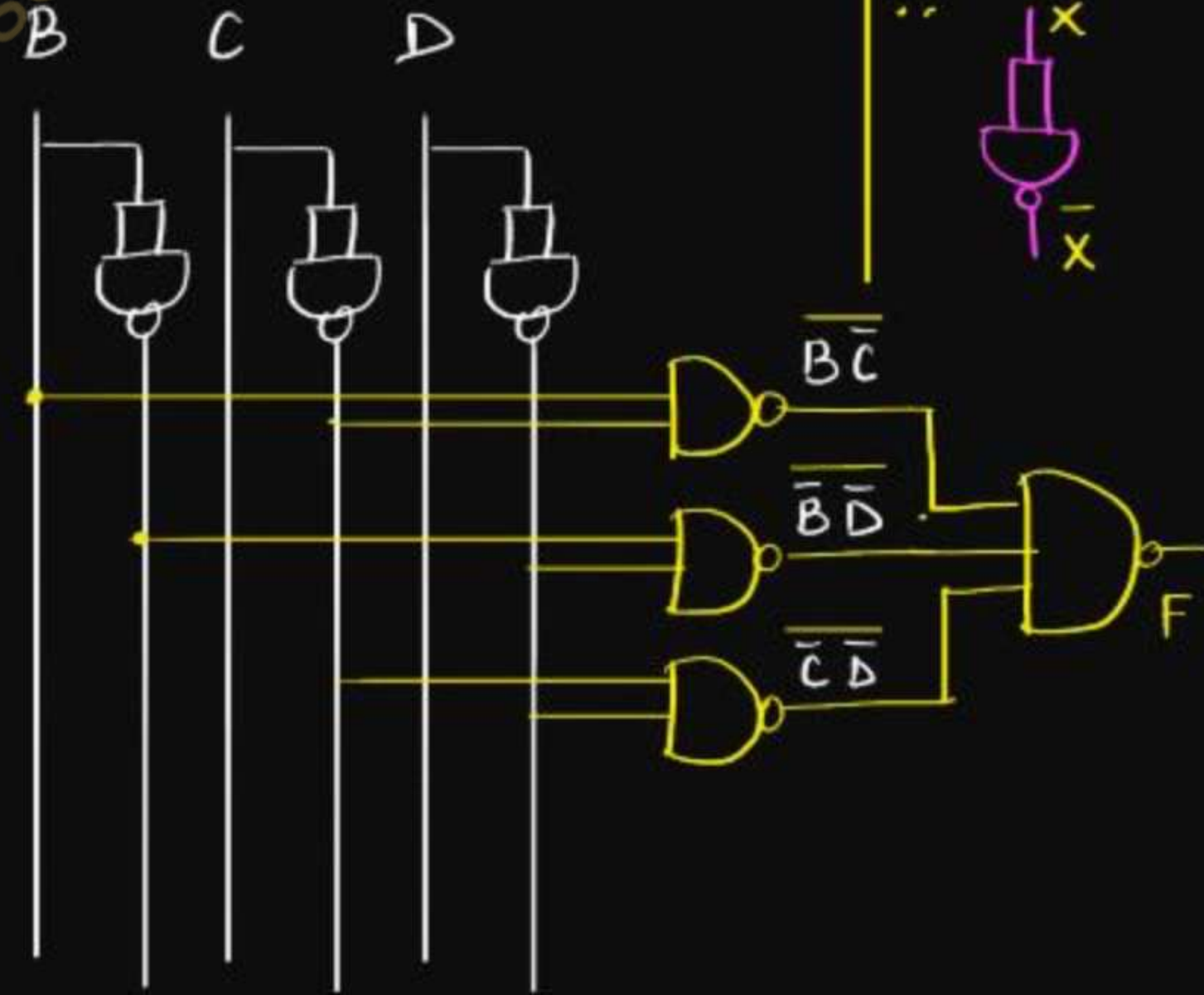
AKTU PYQ-2025-26, Sem III



$$F(A,B,C,D) = B\bar{C} + \bar{B}\bar{D} + \bar{C}\bar{D}$$

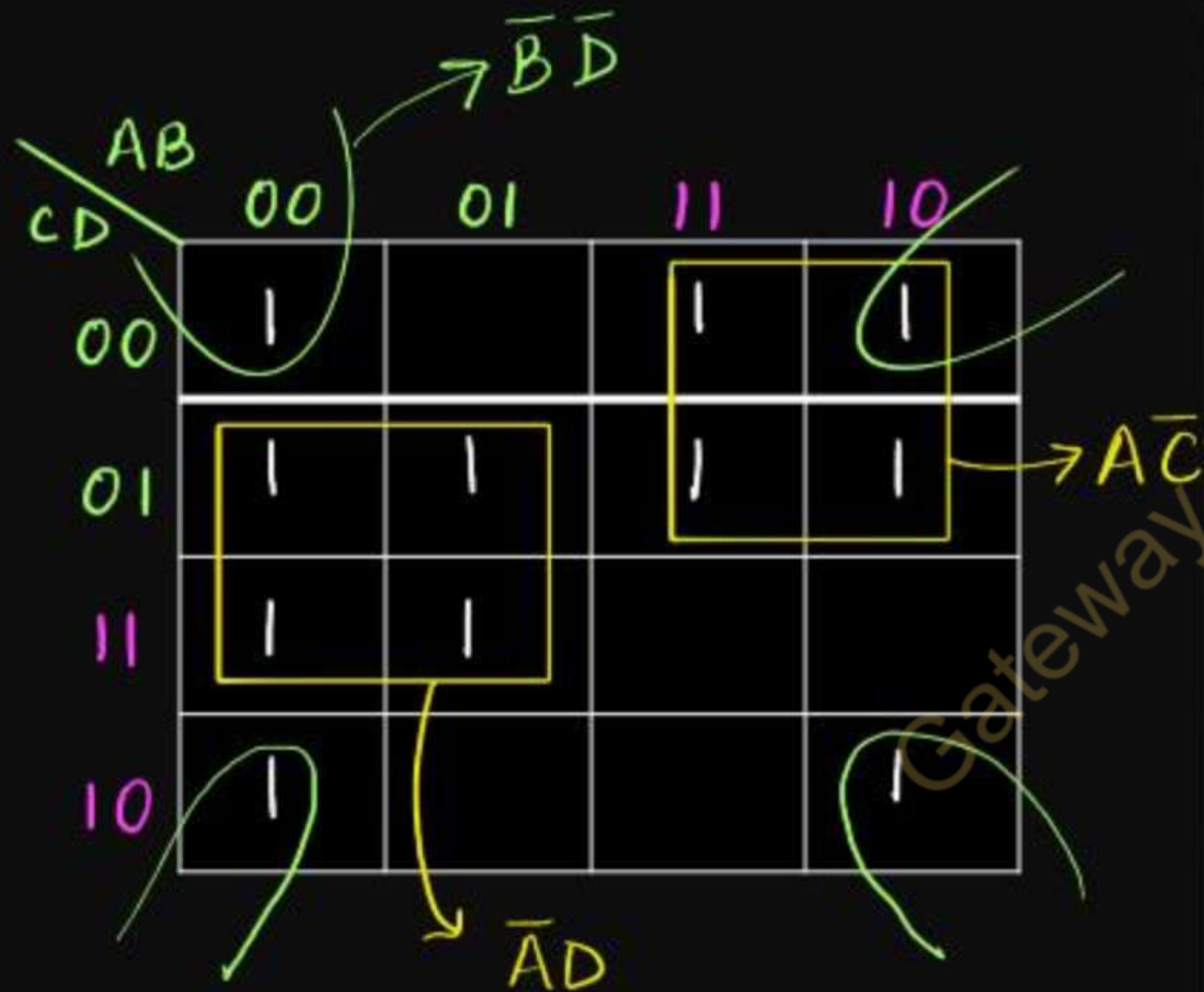
$$F(A,B,C,D) = B\bar{C} + \bar{B}\bar{D} + \bar{C}\bar{D}$$

$$= B\bar{C} \cdot \bar{B}\bar{D} \cdot \bar{C}\bar{D}$$



Que) Reduce the expression $F(A,B,C,D) = \sum m (0,1,2,3,5,7,8,9,10,12,13)$ using K-maps and implement the real minimal expression using NAND logic.

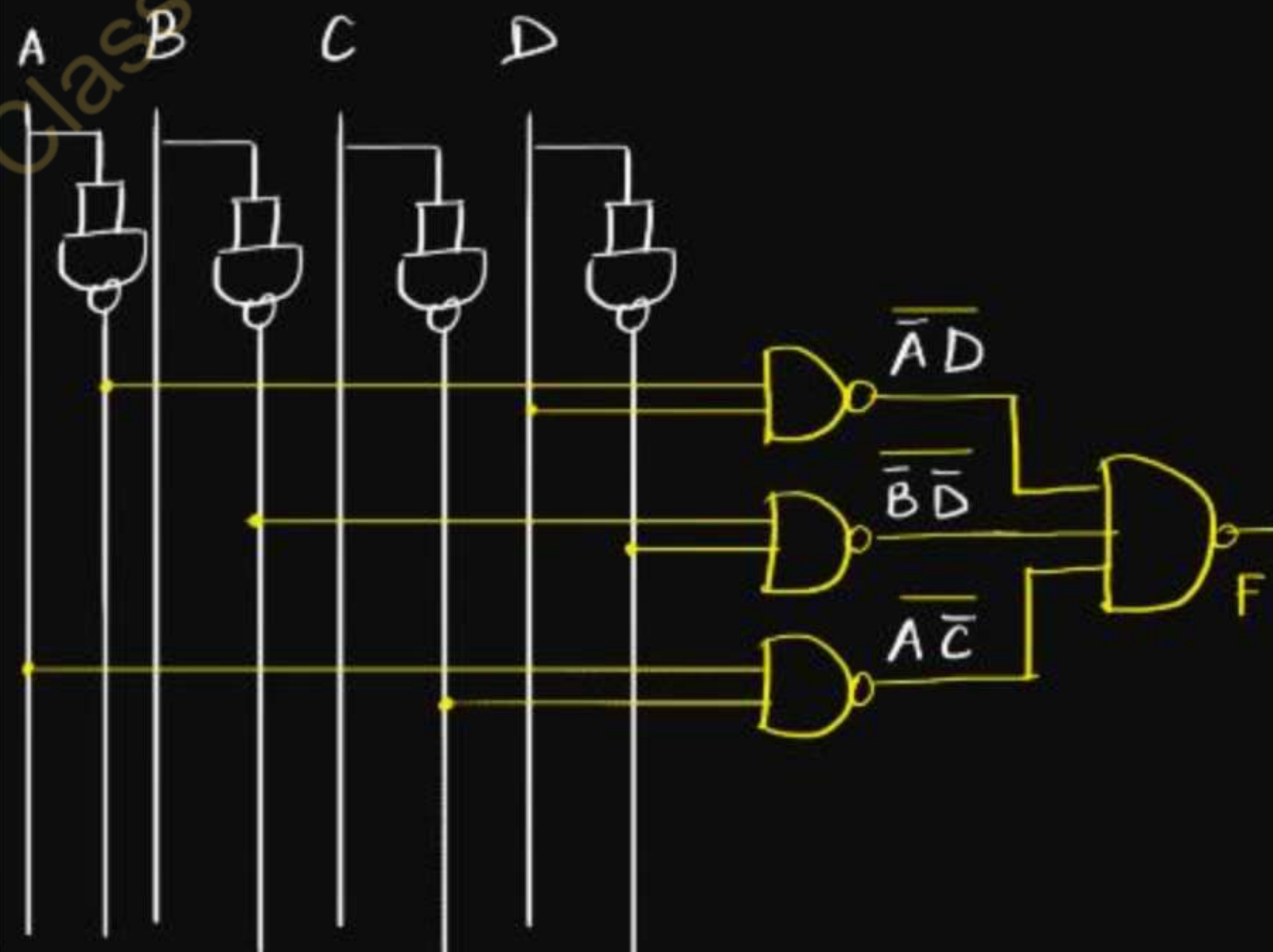
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$$F(A,B,C,D) = \overline{A}D + \overline{B}\overline{D} + AC$$

$$F(A,B,C,D) = \overline{A}D + \overline{B}\overline{D} + AC$$

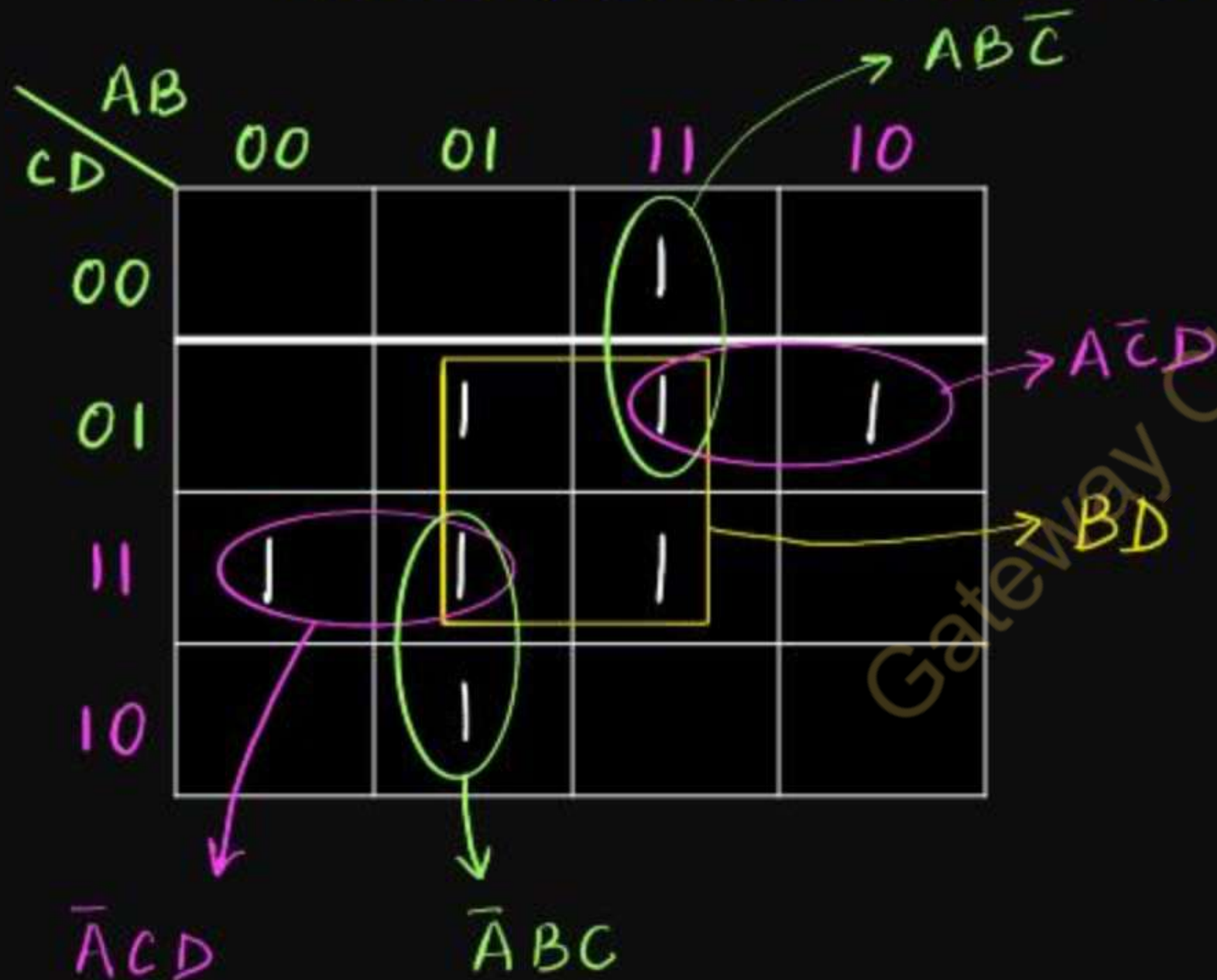
$$= \overline{A}D \cdot \overline{B}\overline{D} \cdot AC$$



Que) Using K-map, simplify the Boolean expression

$$F(A,B,C,D) = \sum m (3, 5, 6, 7, 9, 12, 13, 15)$$

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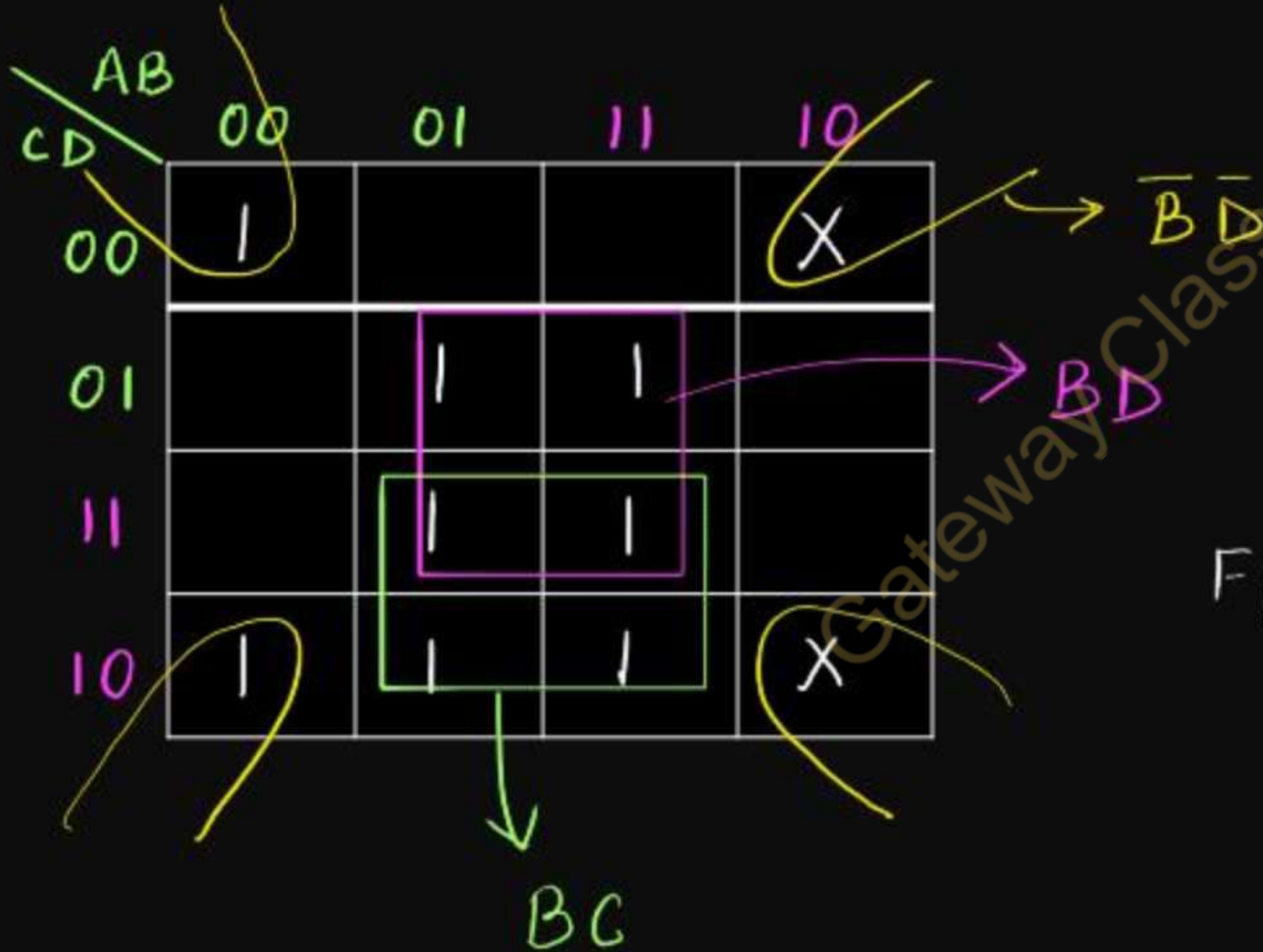
Que) Simplify the Boolean Function $F = \sum m (0,1,2,5,7,8,9,10,13,15) + d (12,14)$ using K-map.

AB \ CD	00	01	11	10
00	1		X	1
01	1	1	1	1
11		1	1	
10	1		X	1

$\bar{B}\bar{D}$ (points to cells 00, 10)
 $\bar{C}D$ (points to cells 01, 11)
 BD (points to cells 01, 11)

$$F = BD + \bar{B}\bar{D} + \bar{C}D$$

Que) Simplify the function $F(A, B, C, D) = \sum m(0, 2, 5, 6, 7, 13, 14, 15) + d(8, 10)$ using K-map.



$$F(A, B, C, D) = BD + \bar{B}\bar{D} + BC$$

Que) Simplify the function $F = \sum m(0, 3, 5, 7, 8, 9, 10, 12, 13) + \sum d(1, 6, 11, 14)$ using K-map.

$AB \backslash CD$	00	01	11	10
00	1		1	1
01	X	1	1	1
11	1	1		X
10		X	X	1

$\bar{A}\bar{D}$ (grouping 00, 01, 11, 10 in column 00)
 $A\bar{C}$ (grouping 00, 01, 11, 10 in row 00)
 $\bar{B}\bar{C}$ (grouping 00, 01, 11, 10 in column 01)
 $A\bar{B}$ (grouping 00, 01, 11, 10 in column 10)

$$F = A\bar{C} + \bar{A}D + A\bar{B} + \bar{B}\bar{C}$$

5 Variable K-Map

(A, B, C, D, E)

$2^5 = 32 \rightarrow 32$ entries in Truth Table

32 cells in K-Map (Cell no. 0 to (n-1) i.e. (0 to 31))

A	B	C	D	E	Cell No.
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	2
0	0	0	1	1	3
0	0	1	0	0	4
0	0	1	0	1	5
0	0	1	1	0	6
0	0	1	1	1	7
0	1	0	0	0	8
0	1	0	0	1	9
0	1	0	1	0	10
0	1	0	1	1	11

A	B	C	D	E	Cell No.
0	1	1	0	0	12
0	1	1	0	1	13
0	1	1	1	0	14
0	1	1	1	1	15
1	0	0	0	0	16
1	0	0	0	1	17
1	0	0	1	0	18
1	0	0	1	1	19
1	0	1	0	0	20
1	0	1	0	1	21
1	0	1	1	0	22
1	0	1	1	1	23

A	B	C	D	E	Cell No.
1	1	0	0	0	24
1	1	0	0	1	25
1	1	0	1	0	26
1	1	0	1	1	27
1	1	1	0	0	28
1	1	1	0	1	29
1	1	1	1	0	30
1	1	1	1	1	31

A = 0

BC DE	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

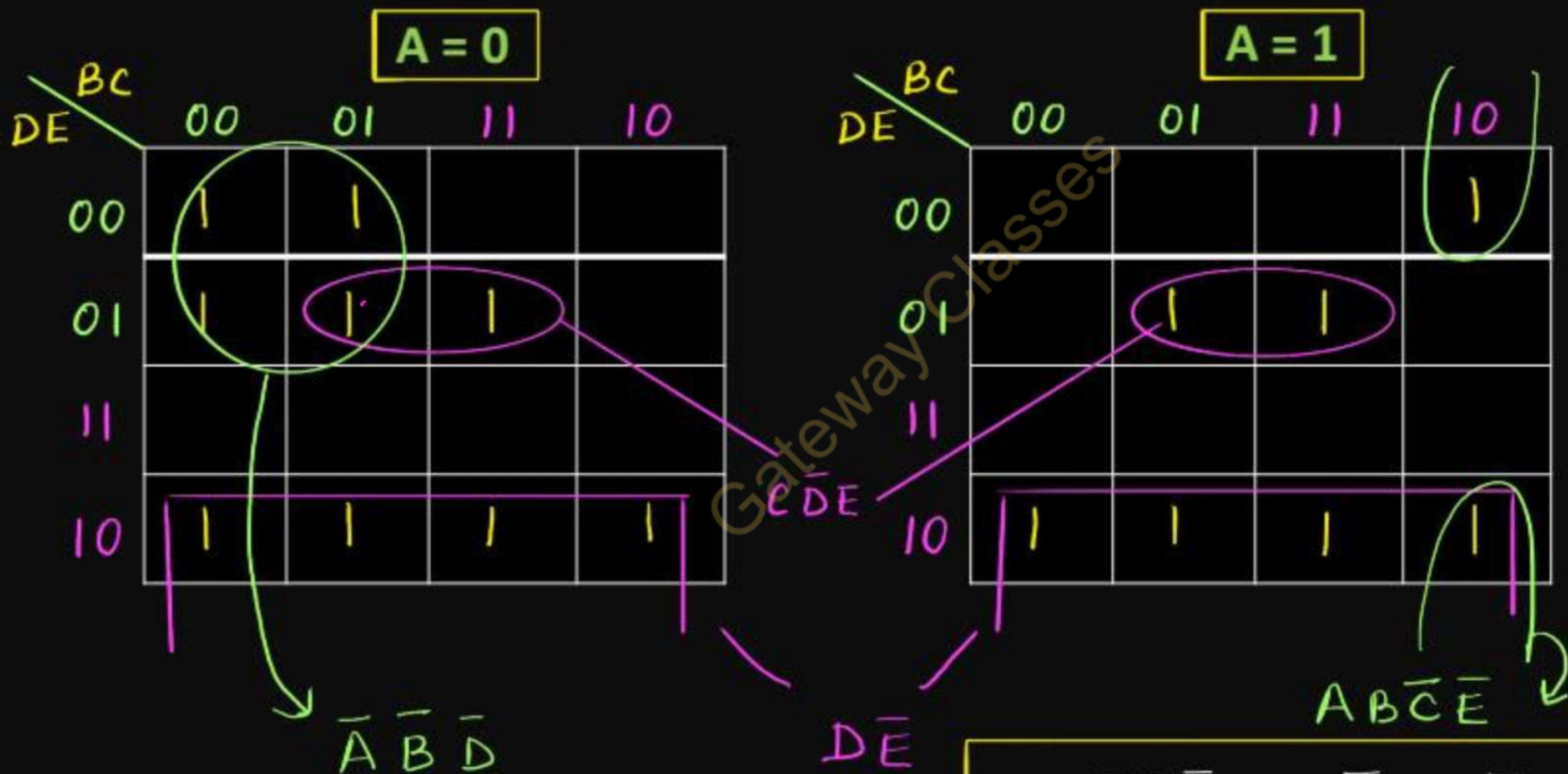
A = 1

BC DE	00	01	11	10
00	16	20	28	24
01	17	21	29	25
11	19	23	31	27
10	18	22	30	26

Steps:-

1. Overlap both K-Maps & Extract answer.
2. When no further overlapping is possible, consider individual K-Map.

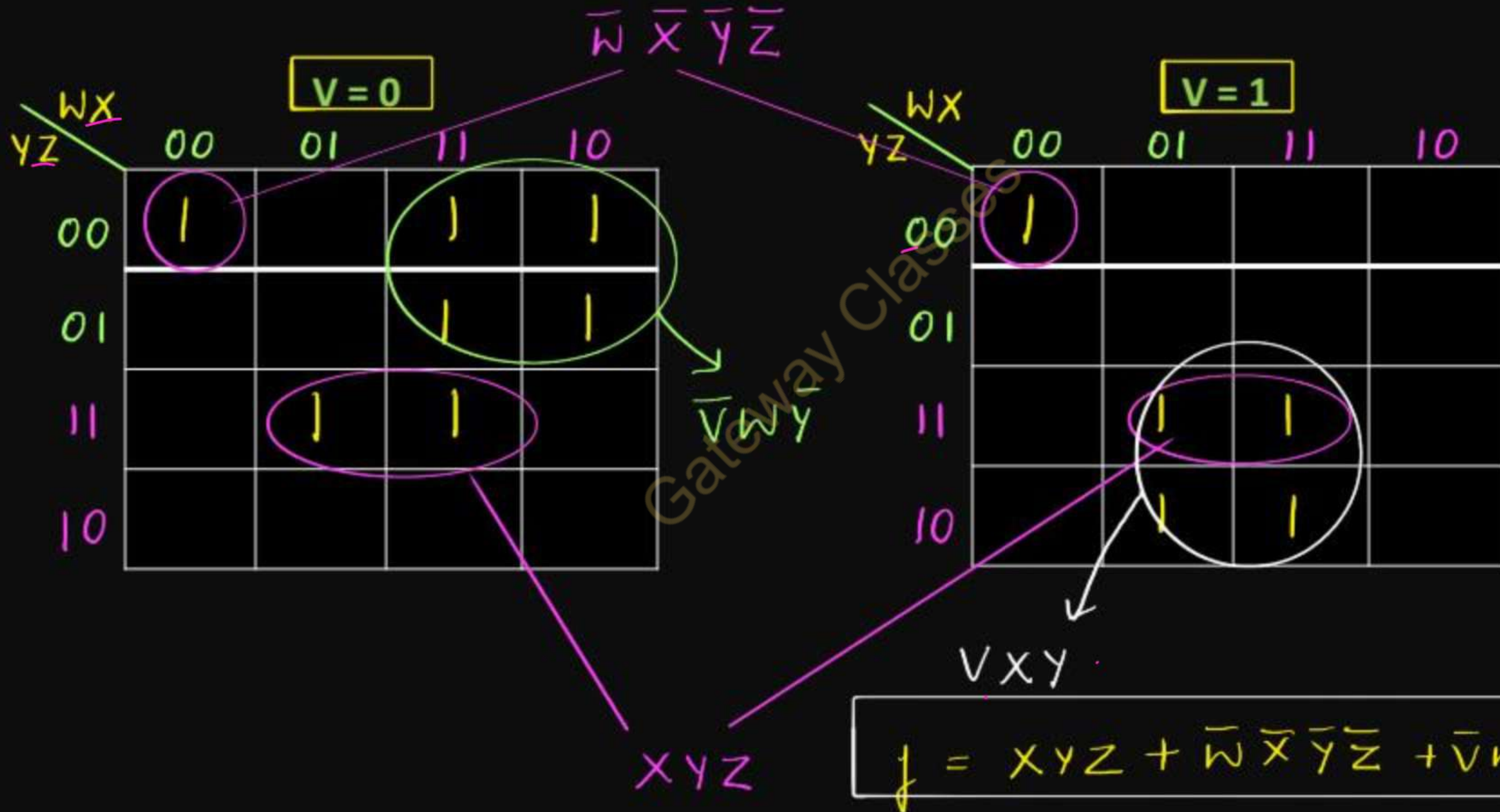
Que) $F(A, B, C, D, E) = \sum m(0, 1, 2, 4, 5, 6, 10, 13, 14, 18, 21, 22, 24, 26, 29, 30)$. Simplify the function with help of K-map.



$$F = \bar{A}\bar{B}\bar{D} + D\bar{E} + C\bar{D}E + AB\bar{C}\bar{E}$$

Que) Simplify the following Boolean expression using K-map.

$$f(V, W, X, Y, Z) = \sum m(0, 7, 8, 9, 12, 13, 15, 16, 22, 23, 30, 31)$$



K-Map for POS Form

Que) $Y = (A + \bar{B})(A + B)(\bar{A} + B)$

1st inv $\begin{matrix} \swarrow \searrow \\ 10 \quad 01 \end{matrix}$ $\begin{matrix} \swarrow \searrow \\ 11 \quad 00 \end{matrix}$ $\begin{matrix} \swarrow \searrow \\ 01 \quad 10 \end{matrix}$

2 Variable K-Map

POS $\rightarrow$ Boolean $\rightarrow$ Double inversion

$\Pi M(-, -) \rightarrow$ Single inversion

A \ B	0	1
0	0	0
1	0	

$\bar{A}$ $\xrightarrow{2^{nd}}$ A

$\bar{B}$ $\xrightarrow{2^{nd}}$ B

$$Y = A \cdot B$$

(Both the answers are multiplied)

Que $Y = \Pi M(0, 1, 3)$

A \ B	0	1
0	0	
1	0	0

$\bar{A}$ $\rightarrow$ A

$B \rightarrow \bar{B}$

$$Y = A \bar{B}$$

Que) $F = \Pi M(0, 1, 3, 4, 5)$

AB \ C	00	01	11	10
0	0			0
1	0	0		0

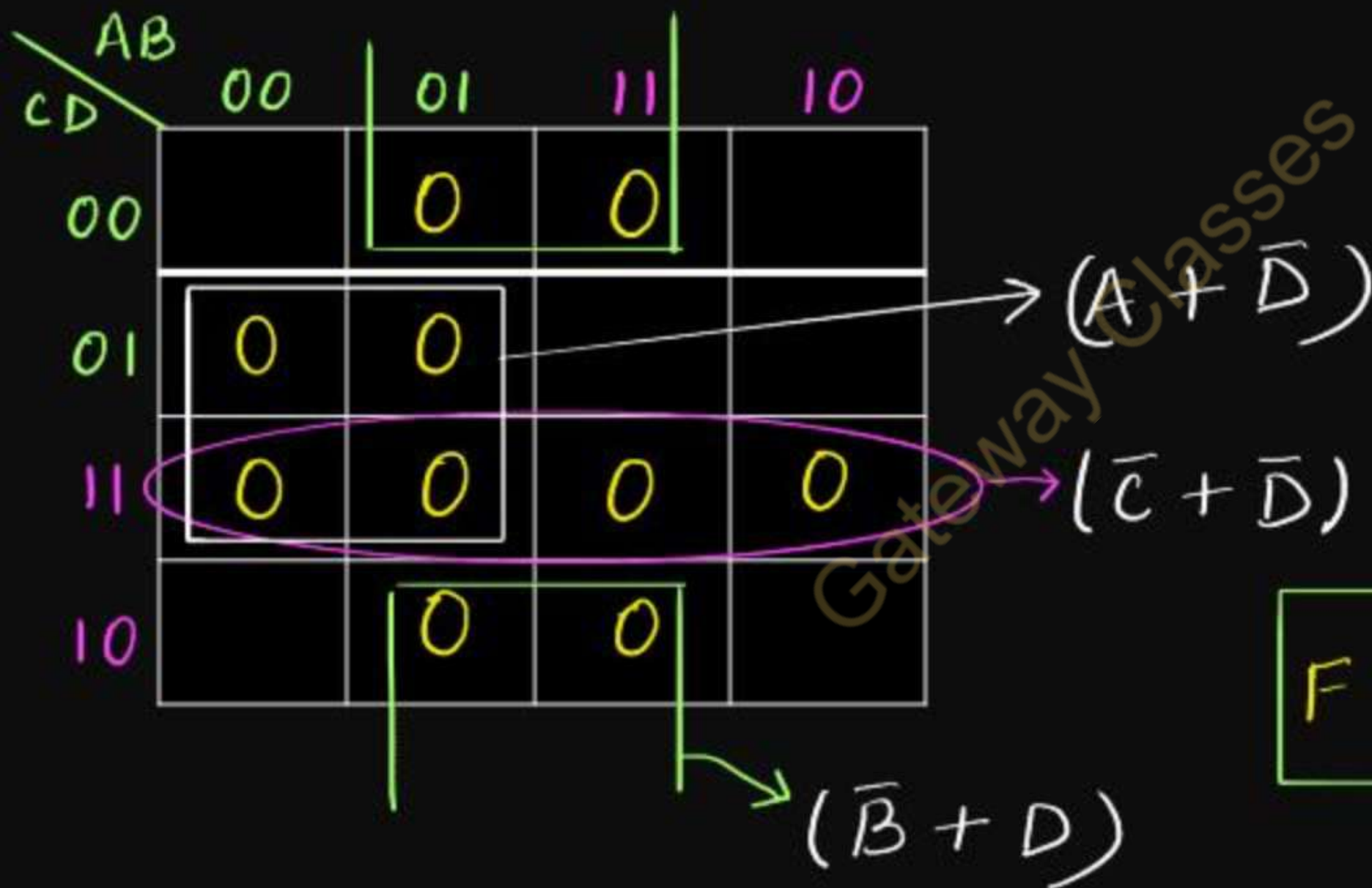
$(A + \bar{C})$

$(A + \bar{C}) \cdot B$

Gateway Classes

4 Variable K-Map

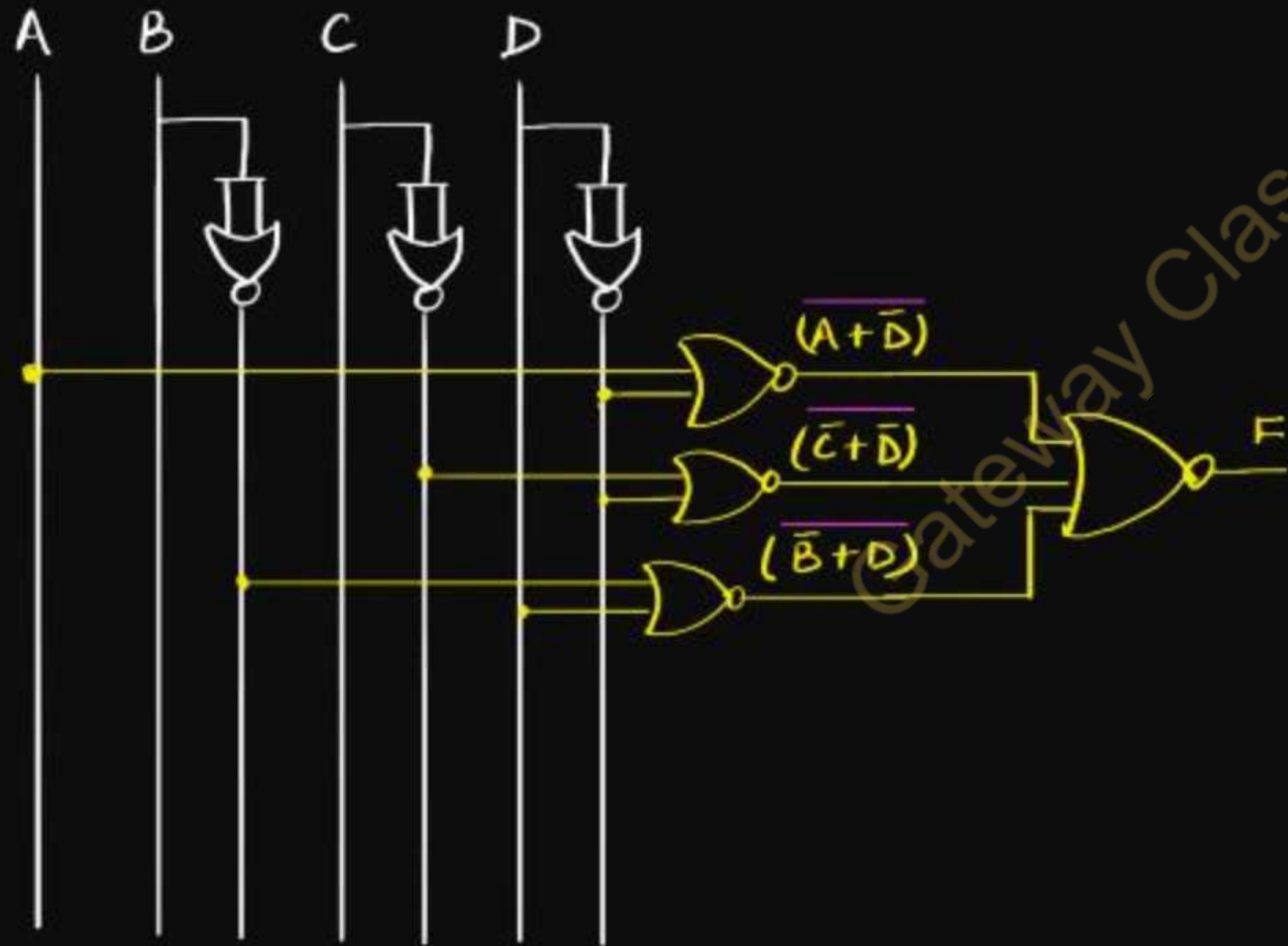
Que) Simplify the Boolean expression using K-map and implement the simplified expression using NOR GATES only $F(A, B, C, D) = \prod M(1, 3, 4, 5, 6, 7, 11, 12, 14, 15)$



$$F = (A + \bar{D})(\bar{C} + \bar{D})(\bar{B} + D)$$

$$F = (A + \bar{D})(\bar{C} + \bar{D})(\bar{B} + D)$$

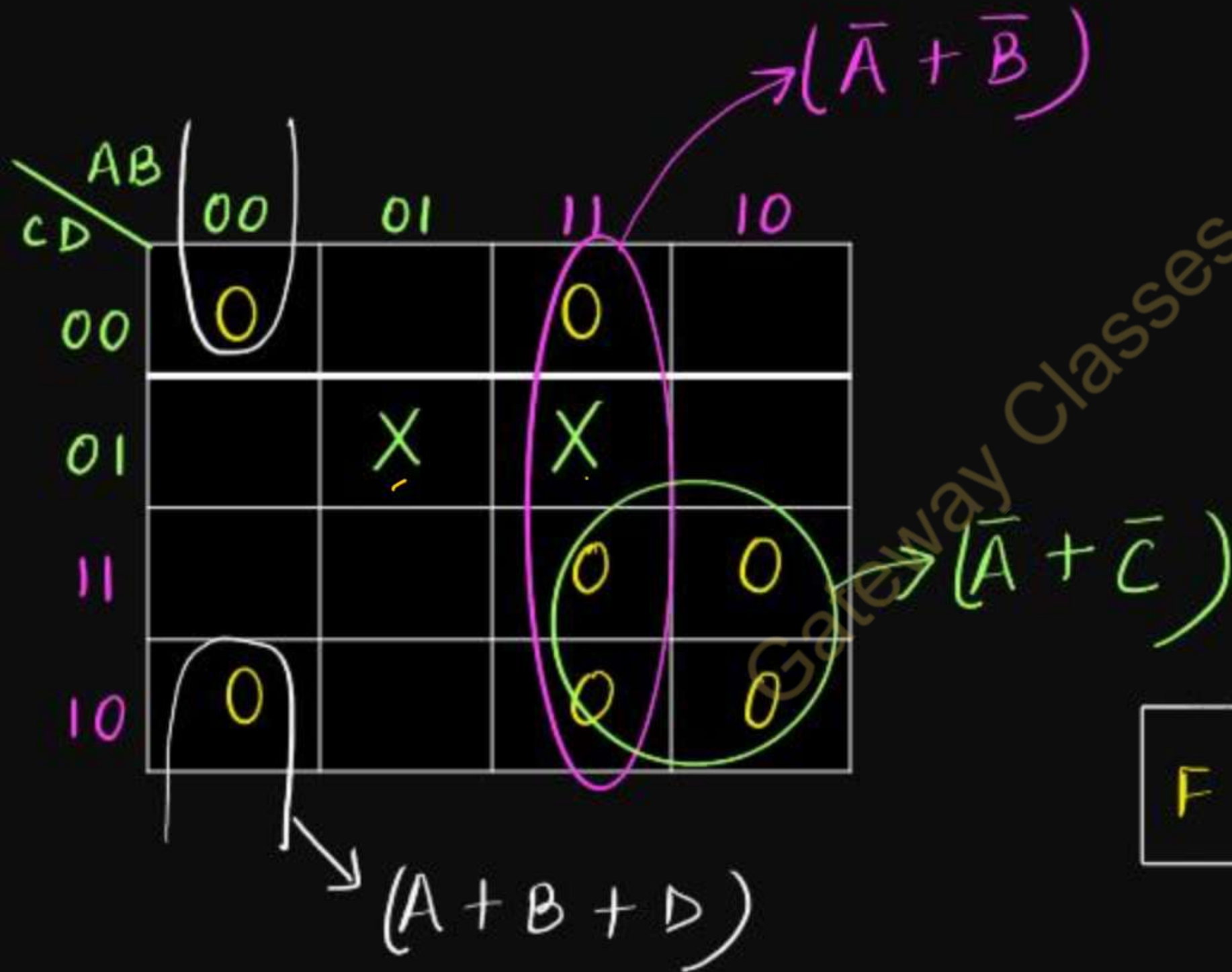
$$(A + \bar{D}) + (\bar{C} + \bar{D}) + (\bar{B} + D)$$



NOR as
NOT



Que) Simplify the function $F = \prod M(0, 2, 10, 11, 12, 14, 15) \cdot \prod d(5, 13)$ using K-map.



$$F = (A + B + D)(\bar{A} + \bar{C})(\bar{A} + \bar{B})$$

Que) Minimize using K-map

$$F(A, B, C, D) = \Pi M(3, 4, 5, 7, 9, 13, 14, 15). d(0, 2, 8).$$

Handwritten K-map for $F(A, B, C, D)$ with don't care conditions $d(0, 2, 8)$.

AB \ CD	00	01	11	10
00	X	0		X
01		0	0	0
11	0	0	0	
10	X		0	

Groupings and resulting prime implicants:

- Group 1: $(A + \bar{B} + C)$ (Circled 0s at (0,1), (1,1), (1,0))
- Group 2: $(\bar{A} + C + \bar{D})$ (Circled 0s at (0,1), (1,1), (1,0))
- Group 3: $(A + \bar{C} + \bar{D})$ (Circled 0s at (1,1), (1,0), (1,1))
- Group 4: $(\bar{A} + \bar{B} + \bar{C})$ (Circled 0s at (1,1), (1,0), (1,1))

Final minimized expression:

$$F = (\bar{A} + \bar{B} + \bar{C})(A + \bar{C} + \bar{D})(A + \bar{B} + C)(\bar{A} + C + \bar{D})$$

TABULATION METHOD OR QM METHOD

Que) Explain different steps associated to Quine Mc Cluskey (Tabular Method of minimizing Boolean Functions.

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Que) Minimize the following Boolean function using tabulation method:

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 5, 6, 7, 8, 14, 15) + d(4, 11, 13)$$

		8	4	2	1
0	0	0	0	0	0
1	0	0	0	0	1
2	0	0	1	0	
3	0	0	1	1	
5	0	1	0	1	
6	0	1	1	0	
7	0	1	1	1	
8	1	0	0	0	
14	1	1	1	0	
15	1	1	1	1	
4	0	1	0	0	
11	1	0	1	1	
13	1	1	0	1	

Grp	Minterm	A	B	C	D
G_0 no 1	m_0	0	0	0	0
	m_1	0	0	0	1
G_1 (one 1)	m_2	0	0	1	0
	m_4^*	0	1	0	0
	m_6	1	0	0	0
	m_3	0	0	1	1
G_2 Two 1	m_5	0	1	0	1
	m_6	0	1	1	0
	m_7	0	1	1	1
G_3 Three 1	m_{11}^*	1	0	1	1
	m_{13}^*	1	1	0	1
	m_{14}	1	1	1	0
G_4 Four 1	m_{15}	1	1	1	1

Gp	Minterm	A	B	C	D
G_0 not	m_0	0	0	0	0
G_1 (one 1)	m_1	0	0	0	1
	m_2	0	0	1	0
	m_4^*	0	1	0	0
	m_8	1	0	0	0
G_2 Two 1	m_3	0	0	1	1
	m_5	0	1	0	1
	m_6	0	1	1	0
G_3 Three 1	m_7	0	1	1	1
	m_{11}^*	1	0	1	1
	m_{13}^*	1	1	0	1
	m_{14}	1	1	1	0
G_4 Four 1	m_{15}	1	1	1	1

①

Only single bit change is allowed

Gp	Minterm pairs	A	B	C	D
G_0	$m_0 - m_1$ $m_0 - m_2$ $m_0 - m_4$ $m_0 - m_8$	0	0	0	—
		0	0	—	0
		0	—	0	0
		—	0	0	0
G_1	$m_1 - m_3$ $m_1 - m_5$ $m_2 - m_3$ $m_2 - m_6$ $m_4 - m_5$ $m_4 - m_6$	0	0	—	1
		0	—	0	1
		0	0	1	—
		0	—	1	0
		0	1	0	—
		0	1	—	0
G_2	$m_3 - m_7$ $m_3 - m_{11}$ $m_5 - m_7$ $m_5 - m_{13}$ $m_6 - m_7$ $m_6 - m_{14}$	0	—	1	1
		—	0	1	1
		0	1	—	1
		—	1	0	1
		0	1	1	—
		—	1	1	0
G_3	$m_7 - m_{15}$ $m_{11} - m_{15}$ $m_{13} - m_{15}$ $m_{14} - m_{15}$	—	1	1	1
		1	—	1	1
		1	1	—	1
		1	1	1	—

②

Put — wherever
bit changes

Gp	Minterm pairs	A	B	C	D
G ₀	m ₀ - m ₁	0	0	0	—
	m ₀ - m ₂	0	0	—	0
	m ₀ - m ₄	0	—	0	0
	m ₀ - m ₈	—	0	0	0
G ₁	m ₁ - m ₃	0	0	—	1
	m ₁ - m ₅	0	—	0	1
	m ₂ - m ₃	0	0	1	—
	m ₂ - m ₆	0	—	1	0
	m ₄ - m ₅	0	1	0	—
	m ₄ - m ₆	0	1	—	0
G ₂	m ₃ - m ₇	0	—	1	1
	m ₃ - m ₁₁	—	0	1	1
	m ₅ - m ₇	0	1	—	1
	m ₅ - m ₁₃	—	1	0	1
	m ₆ - m ₇	0	1	1	—
	m ₆ - m ₁₄	—	1	1	0
G ₃	m ₇ - m ₁₅	—	1	1	1
	m ₁₁ - m ₁₅	1	—	1	1
	m ₁₃ - m ₁₅	1	1	—	1
	m ₁₄ - m ₁₅	1	1	1	—

(3)

— same position are grouped together

Gp	Minterm Quad	A	B	C	D
G ₀	0 - 1 - 2 - 3	0	0	✓	—
	0 - 1 - 4 - 5	0	—	0	—
	0 - 2 - 1 - 3	0	0	—	—
	0 - 2 - 4 - 6	0	—	—	0
	0 - 4 - 1 - 5	0	—	0	—
	0 - 4 - 2 - 6	0	—	—	0
G ₁	1 - 3 - 5 - 7	0	—	—	1
	1 - 5 - 3 - 7	0	—	—	1
	2 - 3 - 6 - 7	0	—	1	—
	2 - 6 - 3 - 7	0	—	1	—
	4 - 5 - 6 - 7	0	1	—	—
	4 - 6 - 5 - 7	0	1	—	—
G ₂	3 - 7 - 11 - 15	—	—	1	1
	3 - 11 - 7 - 15	—	—	1	1
	5 - 7 - 13 - 15	—	1	—	1
	5 - 13 - 7 - 15	—	1	—	1
	6 - 7 - 14 - 15	—	1	1	—
	6 - 14 - 7 - 15	—	1	1	—

(4)

Note:-

Mark X

on m₀ - m₈

in Table 3

m₀ - m₈ has —
at A, and in G₁
there is not — at
A. ∴ m₀ - m₈ is
not involved. X

== same position are grouped together

Grp	Minterm Quad	A	B	C	D
G ₀	0-1-2-3	0	0	—	—
	0-1-4-5	0	—	0	—
	0-2-1-3	0	0	—	—
	0-2-4-6	0	—	—	0
	0-4-1-5	0	—	0	—
	0-4-2-6	0	—	—	0
G ₁	1-3-5-7	0	—	—	—
	1-5-3-7	0	—	—	—
	2-3-6-7	0	—	—	—
	2-6-3-7	0	—	—	—
	4-5-6-7	0	1	—	—
	4-6-5-7	0	1	—	—
G ₂	3-7-11-15	—	—	1	1
	3-11-7-15	—	—	1	1
	5-7-13-15	—	1	—	1
	5-13-7-15	—	1	—	1
	6-7-14-15	—	1	1	—
	6-14-7-15	—	1	1	—

x
x
x
x
x
x
x
x

←

Minterm Octet	A	B	C	D
0-1-2-3-4-5-6-7	0	—	—	—
0-1-2-3-4-6-5-7	0	—	—	—
0-1-4-5-2-3-6-7	0	—	—	—
0-1-4-5-2-6-3-7	0	—	—	—
0-2-1-3-4-5-6-7	0	—	—	—
0-2-1-3-4-6-5-7	0	—	—	—
0-2-4-6-1-3-5-7	0	—	—	—
0-2-4-6-1-5-3-7	0	—	—	—
0-4-1-5-2-3-6-7	0	—	—	—
0-4-1-5-2-6-3-7	0	—	—	—
0-4-2-6-1-3-5-7	0	—	—	—
0-4-2-6-1-5-3-7	0	—	—	—

Note

Note
G₁
G₂
un
grouped
Mark
X in
Table
(5)

(6)

Write given minterms, Don't care is not written

Minterm Octet	A	B	C	D
0-1-2-3-4-5-6-7	0	—	—	—
0-1-2-3-4-6-5-7	0	—	—	—
0-1-4-5-2-3-6-7	0	—	—	—
0-1-4-5-2-6-3-7	0	—	—	—
0-2-1-3-4-5-6-7	0	—	—	—
0-2-1-3-4-6-5-7	0	—	—	—
0-2-4-6-1-3-5-7	0	—	—	—
0-2-4-6-1-5-3-7	0	—	—	—
0-4-1-5-2-3-6-7	0	—	—	—
0-4-1-5-2-6-3-7	0	—	—	—
0-4-2-6-1-3-5-7	0	—	—	—
0-4-2-6-1-5-3-7	0	—	—	—

G₀

Table 3

PI	Minterm Involved	Minterm used											
		0	1	2	3	5	6	7	8	14	15		
$\bar{B}\bar{C}\bar{D}$	0, 8	X							X				
CD	3, 7, 11, 15				X			X			X		
BD	5, 7, 13, 15				X			X			X		
BC	6, 7, 14, 15						X	X		X	X		
$\bar{A}$	0, 1, 2, 3, 4, 5, 6, 7	X	X	X	X	X	X	X					

Table 5

Table 7

Put X in front of min term involved.

Circle ↓ vertically, single in a column.

PI	Minterm Involved	Minterm used									
		0	1	2	3	5	6	7	8	14	15
$\overline{B}\overline{C}\overline{D}$	0, 8	X							X		
CD	3, 7, 11, 15				X			X			X
BD	5, 7, 13, 15					X		X			X
BC	6, 7, 14, 15						X	X		X	X
$\overline{A}$	0, 1, 2, 3, 4, 5, 6, 7	X	X	X	X	X	X	X			

check

$\bar{A}$

Minterm involved
0, 1, 2, 3, 4, 5, 6, 7

$\bar{B}\bar{C}\bar{D}$

0, 8

But given minterms are

0, 1, 2, 3, 5, 6, 7, 8, 14, 15

are not there ↓
add PI which has 14, 15 i.e. BC

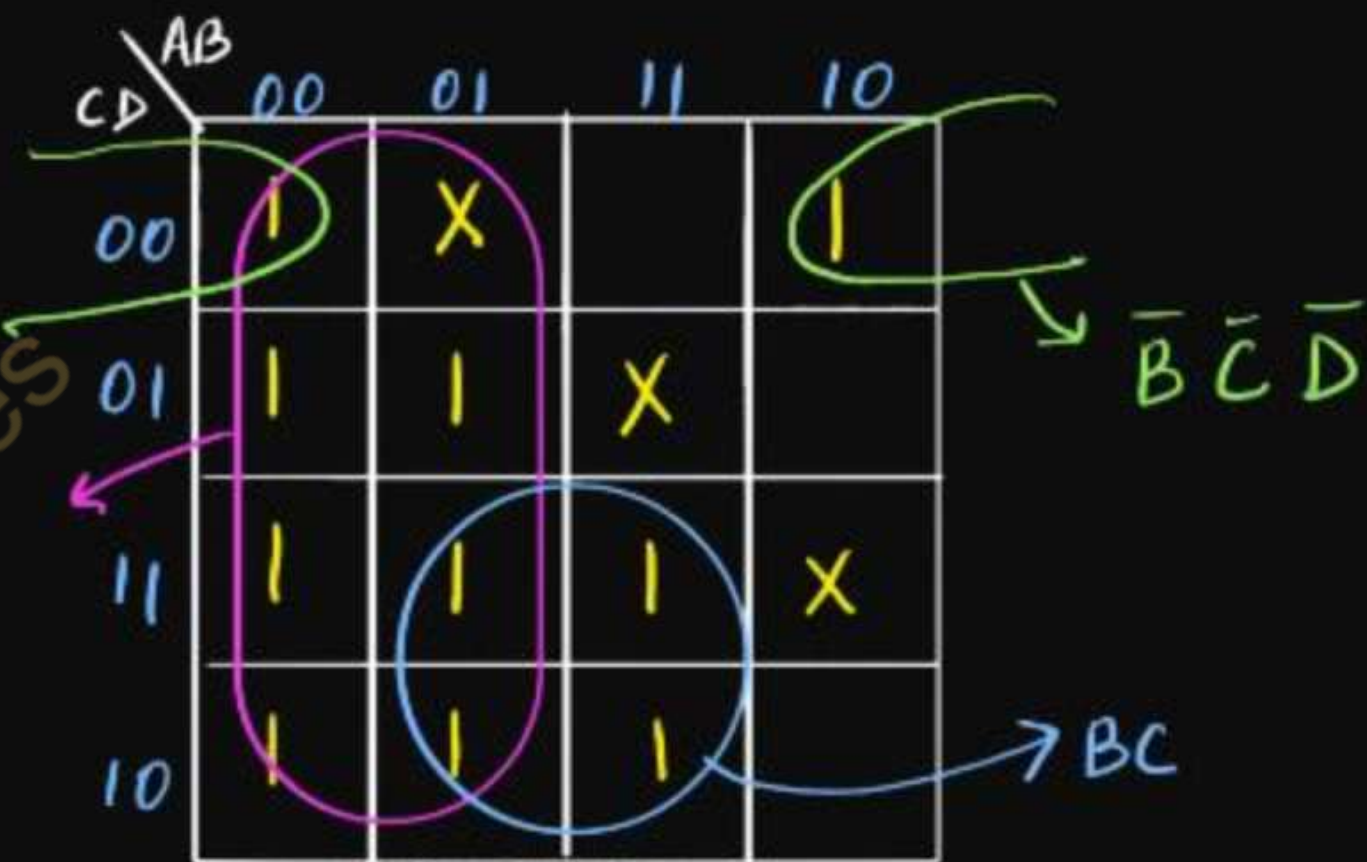
$$F = \bar{A} + \bar{B}\bar{C}\bar{D} + BC$$

Ans

Check by K-Map

$$F(A, B, C, D) = \sum(0, 1, 2, 3, 5, 6, 7, 8, 14, 15) + d(4, 11, 13)$$

AB \ CD	00	01	11	10
00	1	X		1
01	1	1	X	
11	1	1	1	X
10	1	1	1	



$$F = \bar{A} + \bar{B}\bar{C}\bar{D} + BC$$



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